

# Knockout Expression

## The Genotype-to-Expression Strand: the Task, the Model, What the Campaign Established Round by Round, and What to Run Next

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### Abstract

The task is to predict the  $\log_2$  expression ratio of every measured gene in a single-deletion strain from the deletion alone, on the Kemmeren and Sameith compendia, scored by the per-gene Pearson across held-out strains against a replicate ceiling of 0.775. The incumbent, a graph-masked transformer whose encoder runs once on the wild-type graph and whose perturbation enters afterward through a one-key cross-attention, sits at  $0.1965 \pm 0.0222$  across the eight long-budget arms of the v9 mask-schedule round, 25% of the ceiling. Every score in this document is one statistic, the maximum of a centered 5-epoch rolling mean of the validation Pearson, read from the full history of every run and placed against those eight, whose spread bounds the replicate spread from above rather than measuring it. The campaign that followed the SIMB retrospective ran five rounds on IGB and Delta between 2026-08-31 and 2026-09-09.

What is established. The distributional head is not load-bearing: a point head collapses, the CRPS head collapses two times in three, and the Laplace-CRPS head ties the quantile head to 0.0005. Every decoder family tried before the campaign, the GEARS-style cross-gene readout among them, was stopped inside the first 300 epochs and was never compared. The validation loss bottoms out at a few hundred epochs in every run while the metric keeps rising to beyond 19,000, so the long budget buys Pearson on a rising objective. Training on the metric itself does not escape that: pure Pearson at batch 32 collapses to constant outputs at every seed, the batch-64 variant that survives peaks by epoch 2,400 on the long-budget band and gives part of it back, and a ranking objective (ListMLE) learns without collapsing but sits below the band at matched epochs so far. The one factor that moves the score by more than the replicate spread is the content of the gene embedding, ProtT5 over random by 0.07 at 1,000 epochs against a within-cell replicate sd of 0.025; trunk depth, readout width and weight decay do not. The per-gene readout is +0.03 over its in-round reference at two seeds, under the resolution of that round, and the pair term alone is a null. Four of four packed five-day tasks died of host memory.

What is not. Nothing measured beats the incumbent's mean at any budget; the incumbent's own embedding, `calm`, has not been placed against ProtT5; the CLS state is wild-type for every strain and no arm has yet put the perturbation inside the encoder; and no design run so far resolves a gap under about 0.04 at the full budget.

**How to read this.** section 1 is the strand as it stood before the campaign: the task, the model and where strain identity enters it, the heads and decoder families already tried, the published operators, and the imputation signal, carried over from the SIMB retrospective so nothing here needs that document. section 2 is the state of the strand claim by claim, with script, statistic and sample size. section 3 is the readout of the campaign's rounds. section 5 is what the evidence leaves open and what fits the hardware now free.

Working note: `notes/experiments.019-simb-multimodal.expression-strand-retrospective.md`

Generating scripts: `experiments/019-simb-multimodal/scripts/`

Leaderboard: `experiments/019-simb-multimodal/results/round_leaderboards.csv`

Section status: ✓ ready ■ author-reviewed, keep stable ✗ not done, agents may edit freely

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# 1 The strand before the campaign: task, model, and what was already known ✗

## 1.1 The task and its ceiling ✗

The **expression strand** predicts, for a strain carrying one gene deletion, the  $\log_2$  ratio of every measured gene’s transcript abundance against wild type. Targets are the Kemmeren 2014 knockout microarray compendium and the Sameith 2015 double-deletion compendium ([doi:10.1016/j.cell.2014.02.054](https://doi.org/10.1016/j.cell.2014.02.054) and [doi:10.1038/sdata.2015.15](https://doi.org/10.1038/sdata.2015.15)), fused in the **fig3\_core** build: 1,556 records, of which 1,484 are single deletions and 72 are Sameith doubles, so 95.4 % of the training signal carries exactly one perturbed gene. Every deleted gene appears exactly once, so under the strain-level partition no gene is in both the fit and the validation split and every validation strain is an unseen perturbation; a perturbation cannot be represented by its own observed response and must enter through an external gene embedding.

SRC: `experiments/019-simb-multimodal/results/fig3_overlap_census.json`

The score is **pearson\_per\_feature**: for each reporter gene the Pearson correlation across held-out strains between predicted and measured  $\log_2$  ratio, averaged over reporters. A model that predicts each gene’s mean scores exactly zero on it, because a prediction constant across strains has no variance to correlate; the companion per-strain metric, **pearson\_per\_instance**, stays near 0.4 under that same collapse and is the form the perturbation-response literature reports (section 1.4). The **reproducibility ceiling** for the per-feature score is estimated from the 82 deletions measured independently in both compendia, as the mean over reporters of  $\sqrt{\text{clip}(\hat{\rho}_g, 0, 1)}$  with  $\hat{\rho}_g$  the reporter’s test-retest correlation across the paired strains: **0.7746**. The agreement between the two studies’ own measurements of those 82 deletions is 0.611, which is the ceiling for any cross-study statement. SRC: `experiments/019-simb-multimodal/results/expression_ceiling_replicate.json`

**The byte count misleads.** The compendium holds 9.2 million scalar labels against 0.39 million trigenic-interaction labels in the 010 queries, and the intuition that expression would therefore be the easier task was wrong. In the trigenic data a gene recurs across many records with different partners, so the supervision repeatedly constrains its contribution; in the expression data a gene is the perturbation exactly once and is measured on one array, so its 6,169 labels are 6,169 correlated readouts of a single perturbation event and nothing in the data isolates the deletion’s effect from the array’s. The perturbation axis has 1,484 points, and no objective or mechanism changes that.

## 1.2 The model, and the one place strain identity enters ✗

The incumbent is the equivariant cell-graph transformer (`equivariant_cell_graph_transformer.py` under `torchcell/models/`): gene tokens from a frozen embedding (`calm`) projected to width 90, a learnable CLS token prepended, six encoder layers whose self-attention is hard-masked by the nine interaction graphs, then a cross-attention **perturbation operator** whose keys are the deleted genes, then a per-gene readout MLP shared across every gene token. The checkpoint’s state dict reconciles exactly to the logged parameter count.

| module                          | parameters       | role                                                |
|---------------------------------|------------------|-----------------------------------------------------|
| <b>transformer_layers</b>       | 590,220          | 6 graph-masked encoder layers                       |
| <b>embedding_preprocessor</b>   | 369,639          | input projection to $d = 90$                        |
| <b>post_perturbation_mixing</b> | 101,430          | Perceiver mixing after the perturbation             |
| <b>perturbation_transform</b>   | 98,370           | the cross-attention perturbation operator           |
| <b>perturbation_head</b>        | 16,381           | graph-level scalar head                             |
| <b>per_gene_head</b>            | 9,919            | per-gene readout, $d \rightarrow 19$ quantile knots |
| <b>observed_label_encoder</b>   | 8,460            | masked teacher-forcing encoder                      |
| <b>cls_token</b>                | 90               |                                                     |
| <b>total trainable</b>          | <b>1,194,509</b> | matches the logged <b>total_param_count</b>         |

**Table 1:** Parameter budget of the incumbent. The readout row is the striking one: 6,127 reporter genes are predicted by 9,919 parameters, because **PerGeneHead** is one MLP shared across every gene token. The perturbation operator, at 8 % of the parameters, carries the entire burden of making a prediction strain-specific.

**The encoder runs once, on the wild-type graph, and every strain reuses it.** The forward pass encodes the 6,607 gene tokens plus CLS once per batch, with no strain information in the input; the batch dimension appears only at the perturbation operator, which takes the shared gene states  $H$  and each strain’s deleted-gene indices. Two consequences follow directly from the code. The CLS state  $h_{\text{CLS}}$  is wild-type for every strain: measured across-strain standard deviation 0.0 against 0.973 for the pooled perturbed-set summary  $z_S$ , so the conditioner input  $[z_S; h_{\text{CLS}}]$  carries strain identity in one half only, and the FiLM arm was rewired to read  $z_S$  alone because half its first-layer

weights saw a constant. And the perturbation never enters the graph-masked attention at all: the graphs shape how wild-type gene states mix, and the deletion is applied to the finished states afterward. SRC: torchcell/models/equivariant\_cell\_graph\_transformer.py(PerGeneHeadconstructionandtheforwardpass;thesdfiguresarerecordedinthecodecommentthere)

**At one deletion the operator has no pair term.** The cross-attention’s softmax runs over a key set of size one, so its weight is identically 1. Re-drawing  $W_Q$  and  $W_K$  at standard deviation 10 changes the output by exactly 0.0, and 16,200 of the operator’s 32,760 attention parameters are dead at  $|S_b| = 1$ . The model therefore reduces to  $g(h_i + c_b)$ : reporter identity  $i$  and strain identity  $b$  meet once, by addition, and no term depends on the pair (deleted gene, reporter gene). Masked unmasking does not supply one where the score is taken: at reveal step  $k = 0$  every mask bit is zero, the conditioning branch’s inputs are zero as inputs, and the forward pass is identical to the unconditioned model.

SRC: experiments/019-simb-multimodal/results/perturbation\_selector\_degeneracy.json

| mechanism              | pair rank | how one deletion reaches many genes                                                      |
|------------------------|-----------|------------------------------------------------------------------------------------------|
| additive reference     | 0         | it does not: one $c_b$ added to every gene                                               |
| null sink              | 9         | sigmoid gate per head, $\alpha_i$ depends on the querying gene                           |
| graph propagation      | data      | random-walk mass from the deleted gene over the nine graphs, plus a hop-0 self indicator |
| low-rank bilinear      | $r$       | $\sum_j b_{ij}(h_i) a_j(c_b)$                                                            |
| response basis         | $r$       | factored readout, $r$ shared response directions                                         |
| GEARS-style cross-gene | $r$       | pool over genes, low-rank broadcast back, in the decoder                                 |
| per-gene readout       | –         | one output weight per reporter, as GEARS and State have                                  |
| Hadamard / FiLM        | 90        | $h_i \odot (1 + \gamma(c_b))$ , gene and strain multiply                                 |

**Table 2:** Pair-term mechanisms resident in the model file, each gated to be the exact identity at initialization and selectable by config. The rank is what each can express, a derivation and not a measurement; which have been trained, and at what budget, is section 1.3 and section 3.2.

**Rank is not what binds.** The best rank- $r$  approximation of the  $1,482 \times 6,169$  target matrix, scored with the same per-feature Pearson on the matrix it was fit to, reaches 0.7265 at rank 32 and 0.7799 at rank 64, both at the ceiling. A rank-64 pair term has expressive room by a wide margin, so a null from such an arm is about the mechanism or the optimization and not about capacity. After removing each gene’s mean, the residual gene-gene correlation pattern replicates on held-out strains at split-half  $r = 0.8687$  against a permutation null of  $8.5 \times 10^{-5}$ , with effective rank 32.78; that structure is the reason a per-gene independent readout, which emits 6,169 separate marginals, discards most of what the data hold. SRC: experiments/019-simb-multimodal/results/lowrank\_output\_ceiling.json

SRC: experiments/019-simb-multimodal/results/residual\_covariance\_diagnostic.json

### 1.3 Heads and decoder families already trained ✕

Two comparisons are on record from before the campaign, and neither is a comparison.

**Six distributional heads, five of them never past 200 epochs.** The incumbent’s quantile head rests on the table below, read from the leaderboard as it stood when the campaign was designed.

**The decoder and perturbation families, all inside the early dip.** The GEARS-style cross-gene readout, the rank-32 bilinear pair term, Perceiver post-perturbation mixing, the rank-32 response basis, context concatenation, graph propagation, the null sink and the deeper perturbation heads were each trained in waves 1 to 3 of project torchcell\_019\_expr\_v8, most at one seed (table 4). Sixty-seven scored runs across waves 1 to 4b, and not one ran past 300 epochs, against a validation curve that dips at epochs 200 to 300 and then climbs for thousands more (section 2.6). Every one of them was stopped inside the dip, so the differences between arms are early-training transients and more seeds at that budget buy precision on the wrong estimand. Of the families in the table, only two have since been trained at a resolving budget: the rank-64 response basis and the per-gene readout, as the mechanism round (section 3.2). The GEARS-style cross-gene readout, the bilinear term, Perceiver mixing on its own and FiLM have not been.

**Baselines that need no GPU, and one of them is a gate.** Four were run before the campaign’s first round and are the floor every arm is read against.

| dist head    | all runs |        |            | matched budget, 100–200 epochs |        |
|--------------|----------|--------|------------|--------------------------------|--------|
|              | runs     | best   | max epochs | best                           | median |
| quantile     | 344      | 0.2382 | 9,900      | 0.1751                         | 0.1352 |
| crps         | 158      | 0.1631 | 199        | 0.1631                         | 0.1267 |
| energy       | 50       | 0.1421 | 199        | 0.1421                         | 0.0960 |
| laplace_crps | 56       | 0.1411 | 199        | 0.1411                         | 0.0775 |
| point        | 34       | 0.1117 | 163        | 0.1117                         | 0.1054 |
| nll          | 59       | 0.0983 | 199        | 0.0845                         | 0.0680 |

**Table 3:** The head comparison the incumbent rested on, before the objective round. The left block compares one head at 9,900 epochs against five at about 150; the right block is the window where all six ran, on which **quantile** leads by 0.0085 on the median with its maximum drawn from 84 runs against **crps**’s 25. **nll** is the designed  $\sigma$ -collapse control; **energy**’s covariance is global and cannot move a point metric. The objective round (section 2.3) is what replaced this table.

## 1.4 The published operators, and the benchmark that governs the reading ✗

Three papers were read in full against the campaign, mirrored and **sha256**-pinned. They converge on one point.

**Every published operator in this family is additive.** GEARS (Roohani et al. 2023, doi:10.1038/s41587-023-01905-6) combines a perturbation with a gene, and State (Adduri et al. 2025, doi:10.1101/2025.06.26.661135) forms its transformer input, as

$$h_u^{\text{post-pert}} = \text{MLP}(h_u^{\text{gene}} + h^{\mathcal{P}}), \quad \mathbf{H} = \mathbf{H}_{\text{cell}} + \mathbf{H}_{\text{pert}} + \mathbf{H}_{\text{batch}}.$$

CPA (Lotfollahi et al. 2023) is additive latent composition with gene specificity in the decoder. Neither Roohani paper ablates addition against a multiplicative or cross-attentional alternative, so the rank-0 operator of section 1.2 is the field’s default and the structural finding is a statement about the model class.

**Where they differ is the readout.** Both put gene identity in per-gene output parameters applied to a shared nonlinear state: GEARS gives every gene its own linear layer  $w_u \in \mathbb{R}^d, b_u \in \mathbb{R}$ , and State’s readout is one  $\mathbf{W}_{\text{recon}} \in \mathbb{R}^{d_h \times G}$  with a column per gene. The incumbent composes to  $F(h_i + c_b)$  with a single shared  $F$ , so reporter identity enters only through  $h_i$ , inside the same function the perturbation enters. A per-gene output weight costs  $6,127 \times 91 = 557,557$  parameters here, +47%, and it was promoted to the first mechanism arm on this evidence (section 3.2).

**The benchmark.** Ahlmann-Eltze, Huber and Anders 2025 (doi:10.1038/s41592-025-02772-6) benchmarked five foundation models and two purpose-built ones, GEARS and CPA, against deliberately simple baselines: “None outperformed the baselines.” On unseen single perturbations, “None of the deep learning models was able to consistently outperform the mean prediction or the linear model.” Their linear model,  $\arg\min_{\mathbf{W}} \|\mathbf{Y}_{\text{train}} - (\mathbf{G}\mathbf{W}\mathbf{P}^{\top} + \mathbf{b})\|_2^2$ , is B2 above, and it beat GEARS while using GEARS’s own perturbation embeddings. State reports the same of itself on genetic perturbations: “On genetic perturbation datasets, State matched the performance of the perturbation mean baseline”, a 600M-parameter encoder trained on 167 million cells at parity with a mean. Two consequences: scale is not the lever, which the campaign’s own record agrees with at 145 runs from 0.12M to 4.7M parameters and `pearson(log10params, score) = -0.129`; and the “no better than the mean” squared-error result is the field’s central finding stated honestly rather than a local defect. The comparability caveat cuts the other way: their Pearson is computed across genes within one perturbation, where a mean baseline scores high, so the incumbent’s 0.196 and their 0.556 are different statistics and any comparison must use `pearson_per_instance`.

**What is not portable.** State’s contribution is its set machinery, bidirectional attention across sets of 256 cells trained with an energy-kernel MMD, which needs a population of measurements per condition. With one bulk profile per deletion strain, at  $S = 1$  that MMD is  $2\|\hat{x} - x\|_2$ , an ordinary per-sample loss, and cross-cell attention has nothing to attend to. The ceiling rests on 82 cross-study shared deletions, not on per-strain replicates, which is also the argument for replicate measurement in any screen this group designs.

## 1.5 Imputation is the largest signal in the data, and it is orthogonal ✗

A Gaussian conditional mean with a ridge loading, fit once on training strains and never trained by gradient, that observes  $m$  measured genes of a held-out strain and predicts the rest reaches far past anything the model does. At  $m = 0$  it predicts the per-gene mean and scores exactly 0, which is why it is a measurement of information and not a rival model: the model is scored at  $m = 0$ , where the oracle has nothing. SRC: `experiments/019-simb-multimodal/scripts/masked_conditioning_oracle.py`

Knowing which gene was deleted and knowing some of this strain’s other measured genes are almost separate sources of information about the rest, and the strand is scored where only the first is available. The parameter-free companion on

**Table 4:** The decoder and perturbation-family arms of waves 1 to 3 in project `torchcell_019_expr_v8`, 35 runs over 21 arm-wave cells, scored by the maximum of a centered 5-epoch rolling mean of the validation Pearson (`score_decoder_arms.py`). Every run stopped between epochs 50 and 276, inside the early dip of a curve that is still rising at 9,900 (section 2.6), so the scores measure early-training transients and the column is not an arm comparison. The GEARS-style cross-gene readout, the bilinear pair term, Perceiver mixing and the response basis are all here, each at one seed. The mechanism column is a gloss on the override, which is the launcher’s own text.

| Arm             | Wave | Mechanism (override)                                                                                                                                                                                                                                                              | Seeds | Epochs  | Smoothed max, mean / best |
|-----------------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|---------|---------------------------|
| A0_baseline     | 1    | additive reference, $g(h_i + c_b)$ (none)                                                                                                                                                                                                                                         | 3     | 80–120  | 0.081 / 0.153             |
| A1_free16       | 1    | free per-gene lookup of 16 dims<br>( <code>multitask.free_gene_dim=16</code> )                                                                                                                                                                                                    | 1     | 139     | 0.152 / 0.152             |
| A2_self         | 1    | self-indicator only (hop 0) (" <code>\$PROP=true</code> "<br><code>model.perturbation_propagation.hops=0</code> )                                                                                                                                                                 | 1     | 123     | 0.152 / 0.152             |
| A4_prop_h2      | 1    | random-walk propagation, 2 hops<br>( <code>"\$PROP=true"</code> <code>model.perturbation_propagation.hops=2</code> )                                                                                                                                                              | 3     | 70–140  | 0.100 / 0.172             |
| A6_prop_sparse  | 1    | propagation, 2 hops, sparse (" <code>\$PROP=true</code> "<br><code>model.perturbation_propagation.hops=2</code><br><code>"model.perturbation_propagation.graphs=[</code><br><code>regulatory_interaction,tflink,</code><br><code>physical_interaction]"</code> )                  | 1     | 140     | 0.175 / 0.175             |
| B0_protT5       | 1    | ProtT5 gene embeddings (" <code>cell_dataset.</code><br><code>node_embeddings=[prot_T5_all]"</code> )                                                                                                                                                                             | 1     | 70      | 0.089 / 0.089             |
| A6_prop_sparse  | 2    | propagation, 2 hops, sparse (" <code>\$PROP=true</code> "<br><code>model.perturbation_propagation.hops=2</code><br><code>"model.perturbation_propagation.graphs=[</code><br><code>regulatory_interaction,tflink,</code><br><code>physical_interaction]"</code> )                  | 2     | 76–88   | 0.068 / 0.090             |
| C0_concat       | 2    | head sees [ $h_{\text{pert}}; h_i; c_b$ ] ( <code>multitask.</code><br><code>concat_context=true</code> )                                                                                                                                                                         | 1     | 140     | 0.161 / 0.161             |
| D1_bilinear32   | 2    | rank-32 bilinear pair term ( <code>multitask.</code><br><code>bilinear_rank=32</code> )                                                                                                                                                                                           | 1     | 123     | 0.152 / 0.152             |
| E0_perceiver32  | 2    | Perceiver mixing after the perturbation<br>( <code>model.post_perturbation_mixing.enabled=</code><br><code>true</code> <code>model.post_perturbation_mixing.</code><br><code>num_latents=32</code> )                                                                              | 1     | 114     | 0.157 / 0.157             |
| A0_baseline     | 3    | additive reference, $g(h_i + c_b)$ (none)                                                                                                                                                                                                                                         | 1     | 196     | 0.158 / 0.158             |
| A1_ref          | 3    | additive reference, $g(h_i + c_b)$ (none)                                                                                                                                                                                                                                         | 4     | 242–250 | 0.076 / 0.158             |
| A2_prop_sparse  | 3    | propagation, 2 hops, sparse (" <code>\$PROP=true</code> "<br><code>model.perturbation_propagation.hops=2</code><br><code>"model.perturbation_propagation.graphs=[</code><br><code>regulatory_interaction,tflink,</code><br><code>physical_interaction]"</code> )                  | 4     | 50–51   | 0.079 / 0.130             |
| A4_nullsink     | 3    | null-sink gated attention ( <code>model.</code><br><code>perturbation_head.null_sink=true</code> <code>model.</code><br><code>perturbation_head.null_sink_bias_init=-4.</code><br><code>0</code> <code>model.perturbation_head.</code><br><code>null_sink_trainable=true</code> ) | 4     | 186–189 | 0.076 / 0.149             |
| D1_bilinear32   | 3    | rank-32 bilinear pair term ( <code>multitask.</code><br><code>bilinear_rank=32</code> )                                                                                                                                                                                           | 1     | 175     | 0.148 / 0.148             |
| D2_mask         | 3    | hard graph mask replaces the KL ( <code>model.</code><br><code>attention_mask.enabled=true</code> <code>model.</code><br><code>graph_regularization.graph_reg_lambda=0.</code><br><code>0</code> )                                                                                | 1     | 235     | 0.149 / 0.149             |
| E0_perceiver_on | 3    | Perceiver mixing after the perturbation<br>( <code>model.post_perturbation_mixing.enabled=</code><br><code>true</code> <code>model.post_perturbation_mixing.</code><br><code>num_latents=32</code> <code>'model.</code><br><code>post_perturbation_mixing.gate_mode=on'</code> )  | 1     | 181     | 0.127 / 0.127             |
| F1_attn4        | 3    | deeper perturbation head ( <code>model.</code><br><code>perturbation_head.num_layers=4</code> <code>model.</code><br><code>perturbation_head.extra_layer_ffn_mult=0</code> )                                                                                                      | 1     | 222     | 0.155 / 0.155             |
| F1_deep2        | 3    | deeper perturbation head ( <code>model.</code><br><code>perturbation_head.num_layers=2</code> )                                                                                                                                                                                   | 1     | 202     | 0.168 / 0.168             |
| GEARS_crossgene | 3    | GEARS-style cross-gene readout, rank 64<br>( <code>model.cross_gene.enabled=true</code> <code>model.</code><br><code>cross_gene.rank=64</code> )                                                                                                                                  | 1     | 168     | 0.133 / 0.133             |
| H0_factor       | 3    | rank-32 response basis (factored readout)<br>( <code>multitask.response_basis_rank=32</code> )                                                                                                                                                                                    | 1     | 276     | 0.155 / 0.155             |

|    | prediction                                               | what it isolates                          | pf            | nmse   |
|----|----------------------------------------------------------|-------------------------------------------|---------------|--------|
| B0 | each gene’s training mean                                | the definitional floor                    | 0.0000        | 1.0000 |
| B1 | no change, $\log_2$ ratio = 0                            | “nothing happened”                        | 0.0000        | 1.3784 |
| B2 | $\mathbf{GWP}^\top + \mathbf{b}$ , ridge, rank-swept     | a pair term without deep learning         | <b>0.1040</b> | –      |
| B3 | mean of the $k$ nearest deleted genes in embedding space | genotype-side signal from proximity alone | 0.0908        | –      |

**Table 5:** The four baselines, measured 2026-08-31, **pf** = **pearson\_per\_feature**. B2 and B3 are the best over four gene embeddings, both won by **prot\_T5\_all**. B0 and B1 are definitionally zero and B0’s **nmse** of exactly 1.0000 is the quantity **nmse** divides by, so the two are the data-path self-check. The incumbent’s  $0.1965 \pm 0.0222$  clears B2 by 0.093, about 4.2 standard deviations, which is the opposite of what the published benchmark found for GEARS and the reason the campaign’s head question was worth its GPUs. SRC: `experiments/019-simb-multimodal/scripts/expression_baselines.py`

|              | tokens | perturbation enters                                   | per-gene readout                                               |
|--------------|--------|-------------------------------------------------------|----------------------------------------------------------------|
| GEARS        | genes  | $\text{MLP}(h_u + h^P)$                               | yes, $w_u \in \mathbb{R}^d$                                    |
| State        | cells  | $\mathbf{H}_{\text{cell}} + \mathbf{H}_{\text{pert}}$ | yes, $\mathbf{W}_{\text{recon}} \in \mathbb{R}^{d_h \times G}$ |
| torchcell v9 | genes  | $g(h_i + c_b)$                                        | no, one shared MLP                                             |

**Table 6:** Where gene specificity lives in the three operators. These are literature descriptions, not measurements made here.

the genotype side, a similarity-weighted average of the nearest other deleted genes’ phenotypes in **prot\_T5\_all** space, reaches 0.117 against a random-embedding floor of 0.033, so over half the genotype-side signal the trained model finds is available from embedding proximity alone. SRC: `experiments/019-simb-multimodal/results/knn_embedding_probe.json`

## 1.6 The campaign’s arithmetic $\times$

One run is a multi-day object. The v9 **M\_fine** run took 91.4 hours for 9,999 epochs at 32.9 s/epoch and peaked at epoch 9,674; the two IGB card classes deliver the same wall clock, 31.0 and 30.5 s/epoch on RTX 6000 Ada and A100, because validation and checkpointing dominate compute, so 9,900 epochs is 3.5 days solo on either and the partition can be chosen on availability. Packing two runs on one card was measured, not assumed:  $1.53\times$  per-run slowdown for  $1.31\times$  aggregate throughput at 9.5 GiB of VRAM per run, so a packed run reaches about 9,100 epochs in five days. What packing did not measure was host memory, and section 3.6 is the cost of that. SRC: `experiments/019-simb-multimodal/results/expression_objective_diagnosis.json`

SRC: `experiments/019-simb-multimodal/results/launch_plan_evidence.json`

The launch was sized at  $\sigma = 0.0222$ , the spread of the eight long-budget arms, which section 2.1 shows is an upper bound on the replicate spread. At that figure a symmetric two-arm contrast needs 19 runs a side to see 0.02 and 5 to see 0.04, while one fresh arm of three against the eight-run baseline sees 0.042; the matched-budget head gap on record was 0.0085, so no affordable design resolved the real gap and the objective round was built to detect a large effect or report indistinguishability. The replicate spreads measured since, 0.012 to 0.025 within cell at 1,000 epochs and 0.019 at 4,000, make every design in section 5 cheaper than that arithmetic said.

| $m$ revealed | within-study | Kem $\rightarrow$ Sam | % of its $m=1000$ | gain surviving a genotype predictor |
|--------------|--------------|-----------------------|-------------------|-------------------------------------|
| 10           | 0.4201       | 0.2229                | 46 %              | 97.5 %                              |
| 100          | 0.6869       | 0.3668                | 77 %              | 99.2 %                              |
| 1000         | 0.7977       | 0.4793                | 100 %             | 100.6 %                             |

**Table 7:** The masked-conditioning oracle, **pearson\_per\_feature** on validation strains, genes revealed at random. The cross-study column is the honest one, since within-study conditioning partly predicts the target array’s own noise; against the 0.611 cross-study ceiling, 100 genes recover 60 %. The last column replaces the per-gene mean with a  $k$ -nearest-neighbor genotype predictor over **prot\_T5\_all** and re-runs the oracle on its residuals: the gain does not shrink, so the two channels are nearly non-overlapping. Choosing the revealed genes by across-strain residual variance beats random at the cheap end by +0.046 at  $m = 10$  across studies.

## 2 The state of the strand, claim by claim ✗

The task is the one of section 1.1, scored by the per-gene Pearson across held-out strains and averaged over genes against a replicate ceiling of 0.775. Everything below is read from the leaderboard puller (`pull_round_leaderboards.py`) over project `torchcell_019_expr_v9`, which is the masked-label objective, or from a script named in the paragraph. The statistic throughout is `roll_max`: the maximum of a centered 5-epoch rolling mean of `val/expression/pearson_per_feature`, an upward-biased order statistic whose bias grows with the epochs run, and which cancels in a contrast between runs scored the same way.

### 2.1 The incumbent and its spread ✗

The incumbent is the quantile head under the pinball loss, learning rate  $3 \times 10^{-4}$ , dropout 0.1, six transformer layers at hidden width 90, the nine-graph attention mask, `calm` gene embeddings, batch 32, 9,900 epochs, and the reveal schedule `[0, 10, 100, 1000]`. Eight runs at that budget share every one of those settings the leaderboard records and score  $0.1965 \pm 0.0222$ ; they were read as identical-config replicates until 2026-09-08, when their W&B configs showed them to be the eight arms of the v9 mask-schedule round (`M_sched`, the incumbent schedule; `M_lo`, `M_hi`, `M_fine`, `M_coarse`, other schedules; `M_nomix`, mixing off; `M_gate_rezero`, gate closed; `M_off`, no schedule). So  $0.1965 \pm 0.0222$  is a mean and spread across arms, the spread bounding nondeterminism from above; the incumbent schedule itself is one draw at 0.1824, and the three highest leaderboard entries, 0.238 to 0.243, are the `M_fine` arm and its resumed lineage to 19,000 epochs, one draw of one schedule. Replicate spread measured on runs that do share a configuration is 0.0246 pooled within cell at epochs  $\leq 990$  (section 3.1) and 0.019 between two seeds of the mechanism-round reference at epochs  $\leq 4,079$  (section 3.2). Baselines that need no GPU (`expression_baselines.py`): per-gene mean 0.0000, bilinear on ProtT5 0.1040, embedding-neighbor 0.0908. The incumbent clears the bilinear by 0.093, about 4.2 standard deviations. On squared error no live run beats the per-gene mean at its Pearson peak: the best `nmse` at peak is 1.005.

### 2.2 The ceiling of every panel, from its own reproducibility ✗

The per-gene Pearson a perfect predictor can reach is bounded by how reproducibly the panel measures each gene. Write one measurement of gene  $g$  in strain  $b$  as  $x_{bg} = s_{bg} + e_{bg}$ , signal and noise with variances  $\sigma_{s,g}^2$  and  $\sigma_{e,g}^2$  across strains. A predictor that recovers  $s$  exactly scores

$$c_g = \text{corr}(s_{\cdot g}, x_{\cdot g}) = \sqrt{\frac{\sigma_{s,g}^2}{\sigma_{s,g}^2 + \sigma_{e,g}^2}} = \sqrt{\rho_g}, \quad \text{ceiling} = \frac{1}{|G|} \sum_{g \in G} \sqrt{\rho_g}, \quad (1)$$

with  $\rho_g$  the reliability of gene  $g$  and the average taken over genes as the metric takes it. Two estimators of  $\rho_g$  are available, and the panels differ in which one their design allows. With two independent measurements  $x$  and  $y$  of the same strains, the test-retest correlation across strains estimates  $\rho_g$  directly,  $\hat{\rho}_g = \text{corr}(x_{\cdot g}, y_{\cdot g})$ , since  $\text{corr}(s + e, s + e') = \sigma_s^2 / (\sigma_s^2 + \sigma_e^2)$ . With replicate measurements of one genotype but no paired panel, the noise variance comes from the replicates and the total variance from the panel,  $\hat{\rho}_g = 1 - \sigma_{e,g}^2 / \sigma_g^2$  (the decomposition). Both clip  $\hat{\rho}_g$  to  $[0, 1]$  before the square root; a negative estimate is a gene whose replicate disagreement exceeds its spread across the panel and counts as zero. table 8 gives every panel both ways where the data admit it, next to what each paper reports about its own reproducibility.

SRC: `experiments/019-simb-multimodal/results/expression_ceiling_all.json`

**Kemmeren and Sameith.** No replicate correlation is reported by either paper; both state the design, two cultures in dye swap for Kemmeren and four profiles per mutant for Sameith. The 82 deletions both studies profiled are the test-retest pair: per-gene  $r$  of 0.611 across those strains, ceiling 0.775 by eq. (1), 0.862 by the decomposition, the numbers the strand has scored against throughout (`expression_ceiling_replicate.py`).

**Caudal.** The paper’s 0.94 is a per-sample correlation across genes between the two cultures of an isolate, which the mirrored replicate table reproduces at 0.947 over 29 isolates; it says the profiles are the same shape, not that each gene’s variation across isolates is reproducible. Per gene across the 29 pairs the test-retest  $r$  averages 0.40 (median 0.42), a ceiling of 0.60; the decomposition against the 943-isolate variance gives 0.46, with 17% of genes at a negative reliability, so the two estimators disagree more here than on Kemmeren, and the 29 pairs are the reason: a per-gene correlation over 29 points has a standard error near 0.18, and the noise variance of a gene from 29 differences is no better determined. The isolate panel’s ceiling is therefore 0.5 to 0.6, known to about  $\pm 0.1$ .

**Messner.** One proteome per strain, so no replicate pair exists; the panel reports CVs of 8.1% technical, 11.3% across the 388 wild-type replicates and 16.2% across knockouts. The decomposition takes the single-measurement noise to be the wild-type replicate variance per protein, carried on the log2 scale in the converted records, against the variance of the log2 ratio across the 4,476 deletions: mean reliability 0.50, ceiling 0.70, and the paper’s own CVs give the same

**Table 8:** The per-gene Pearson a perfect predictor could reach on each panel, from that panel’s own reproducibility.

Test-retest  $r$  is the per-gene Pearson across strains between two independent measurements of the same strains, averaged over genes; the ceiling is the mean over genes of  $\sqrt{r_g}$  (test-retest) or of  $\sqrt{1 - \sigma_{e,g}^2/\sigma_g^2}$  (decomposition, noise variance from replicates of one genotype, total variance across the panel), reliabilities clipped to  $[0, 1]$ . Messner’s test-retest pairs are strains of different origin carrying the same deletion, not replicate cultures, so that column is a lower bound; its decomposition takes the single-measurement noise to be the wild-type replicate variance. Nadal-Ribelles’ same-genotype agreement across batches is read against the median of different genotypes. SRC: `experiments/019-simb-multimodal/scripts/expression_ceiling_all.py`

| panel                                              | replicate design             | reported                    | test-retest $r$ (ours)                  | ceiling, retest  | test- ceiling, decomposition |
|----------------------------------------------------|------------------------------|-----------------------------|-----------------------------------------|------------------|------------------------------|
| Kemmeren 2014 (mRNA, 1,484 deletions)              | 2 cultures $\times$ dye swap | no $r$ reported             | 0.61 (82 deletions shared with Sameith) | 0.77             | 0.86                         |
| Sameith 2015 (mRNA, 82 + 72 deletions)             | 4 profiles per mutant        | no $r$ reported             | 0.61 (the same pair)                    | 0.77             | –                            |
| Caudal 2024 (mRNA, 943 isolates)                   | 29 isolates re-cultured      | 0.94 per sample (ours 0.95) | 0.40 (29 pairs)                         | 0.60             | 0.46                         |
| Messner 2023 (protein, 4,476 deletions)            | none; 388 WT replicates      | CV 11.3% WT, 16.2% KO       | 0.20 (146 duplicated-origin ORFs)       | 0.40             | 0.70 (CV check 0.72)         |
| Zelezniak 2018 (protein, 97 kinase deletions)      | 2 to 7 cultures per strain   | CVs as a figure             | –                                       | –                | 0.92                         |
| Nadal-Ribelles 2025 (single cell, 2,243 deletions) | batches, no replicate design | no $r$ reported             | 0.04 across batches (null 0.02)         | 0.21 (null 0.15) | –                            |

answer,  $\sqrt{1 - (0.113/0.162)^2} = 0.72$ . The 146 ORFs present as strains of different origin are a harsher test, since they differ in origin and not only in culture: per protein  $r$  of 0.20, ceiling 0.40. The proteome’s ceiling lies between 0.40 and 0.70, the first a lower bound that includes strain-of-origin differences and the second the bound a true replicate would set if knockout noise equals wild-type noise.

**Zelezniak.** Two to seven cultures per kinase deletion with a per-strain standard error stored; the target is the replicate mean, whose noise variance is  $se^2$ , and the decomposition gives a ceiling of 0.92 over 726 proteins. That the two proteomes agree at 0.08 per deletion (section 3.8) while each is reproducible within itself says the disagreement is between the studies, not within either.

**Nadal-Ribelles.** The paper validates against Kemmeren by the count of differentially expressed genes per genotype and reports no genotype-level replicate correlation. Same-genotype pseudobulk profiles measured in different batches correlate at a median 0.043 against 0.022 for different genotypes (`nadal_replication_coverage.py`), a ceiling of 0.21 against a null of 0.15; the panel does not reproduce its own genotype signal above chance, which is why it is not a target.

**What the ceilings say about the campaign.** The incumbent’s  $0.1965 \pm 0.0222$  on Kemmeren is a quarter of that panel’s ceiling. The proteome’s ceiling is lower and less certain, 0.40 to 0.70, and its linear floor on the same partitions is 0.06 to 0.09 (section 4), so a proteome score has less room above the linear model than an expression score does; the v14 round (section 5) reads against both bounds. The isolate panel sits at a ceiling of 0.5 to 0.6 with 943 strains and a perturbation set of thousands of variants per isolate, a different task from a single deletion.

## 2.3 The head is not load-bearing $\times$

The objective round (IGB, launched 2026-08-31, jobs 2368333, 2368337, 2368339; three replicates per challenger at the incumbent’s budget) asked whether the quantile head does anything a point head or another proper scoring head would not.

What the head buys is not collapsing. Between the two heads that do not collapse the choice was never load-bearing, and the round was sized so that only a large effect could have shown; the honest statement is that the heads are indistinguishable at any budget that can be afforded. This is the head question; the decoder-family question, whether a GEARS-style cross-gene readout, a bilinear pair term or Perceiver mixing changes the score, has never been asked at a resolving budget (section 1.3, table 4), and the one family that has, the per-gene readout, is section 3.2.

What the incumbent’s head optimizes and what the metric reads are different things (fig. 1). The head emits 19 quantile knots per gene on the grid  $\tau = 0.05, 0.10, \dots, 0.95$ ; each knot has its own pinball term  $\rho_\tau(u) = \max(\tau u, (\tau - 1)u)$  on its own residual  $u = y - \hat{q}_\tau$ , and the objective is their mean. `pearson_per_feature` reads the median knot alone. On one illustrative gene the median term is 8.1 % of the loss, so the other 18 knots carry 91.9 % of the gradient through the shared trunk toward a quantity the metric never inspects. The knots are 19 points on the predictive CDF, so they also carry a per-gene noise estimate and a calibration check: the fraction of observations that fall inside the central 50 % and 80 % intervals. At epoch 10,000 the incumbent reads 0.326 and 0.578 against 0.50 and 0.80, both far below the diagonal, and both back-solve under a Gaussian scale mismatch to the same predicted-over-true width ratio, 0.624

| head         | $n$ | collapsed | live mean | live sd | live max | min nmse at peak | epochs               |
|--------------|-----|-----------|-----------|---------|----------|------------------|----------------------|
| point        | 6   | 6         | –         | –       | –        | –                | 2,039–9,895          |
| crps         | 6   | 4         | 0.1697    | 0.0094  | 0.1764   | 1.075            | 1,995–9,896          |
| laplace_crps | 6   | 0         | 0.1947    | 0.0167  | 0.2154   | 1.025            | 1,995–9,896          |
| quantile     | 20  | 0         | 0.1960    | 0.0253  | 0.2430   | 1.005            | 1,593–18,990         |
| pearson      | 5   | 3         | 0.1851    | 0.0067  | 0.1899   | 1.365            | 3,162–9,107, running |
| pearson_mse  | 3   | 2         | 0.1859    | –       | 0.1859   | 1.046            | 3,162–6,573, running |
| listmle      | 4   | 0         | 0.1508    | 0.0116  | 0.1676   | 74.7             | 1,789–2,430, running |

**Table 9:** The objective round (upper block) and the two metric-aligned rounds that followed it (lower block), on the leaderboard’s rule, which reads a 500-sample history and so places each score slightly below the full-history readouts of section 3. Collapse is a final Pearson numerically zero: the network emits one constant per gene. The point head collapses every time, CRPS two times in three, the Laplace-CRPS and quantile heads never. Between the two that survive the gap is 0.0005 against a detectable gap of about 0.042 at three replicates versus the eight long-budget arms at their spread of 0.0222, an upper bound on the replicate spread. The lower block is partial: every one of those jobs was still running on IGB at the read, and the two `listmle_mse` runs that collapsed and were killed are below the 1,000-epoch filter. `listmle`’s nmse of 74.7 is not a defect of the fit: the ranking objective is shift-invariant, so each gene’s predicted location drifts freely while its ordering, which the metric reads, does not.

and 0.627: the intervals are  $1.6\times$  too narrow. Simulating that one ratio reproduces the two measured coverages. The spread that most of the loss is spent on is itself miscalibrated, in the overconfident direction, while the point estimates are over-dispersed at  $s/r = 1.92$  (SIMB retrospective, campaign section).

## 2.4 Spread across the long-budget arms is a function of budget, and not a monotone one x

The same eight long-budget arms, rescored on the prefix of their own curves (`short_budget_spread.py`). Read every spread below as an arm spread plus nondeterminism, an upper bound on the replicate spread at that budget:

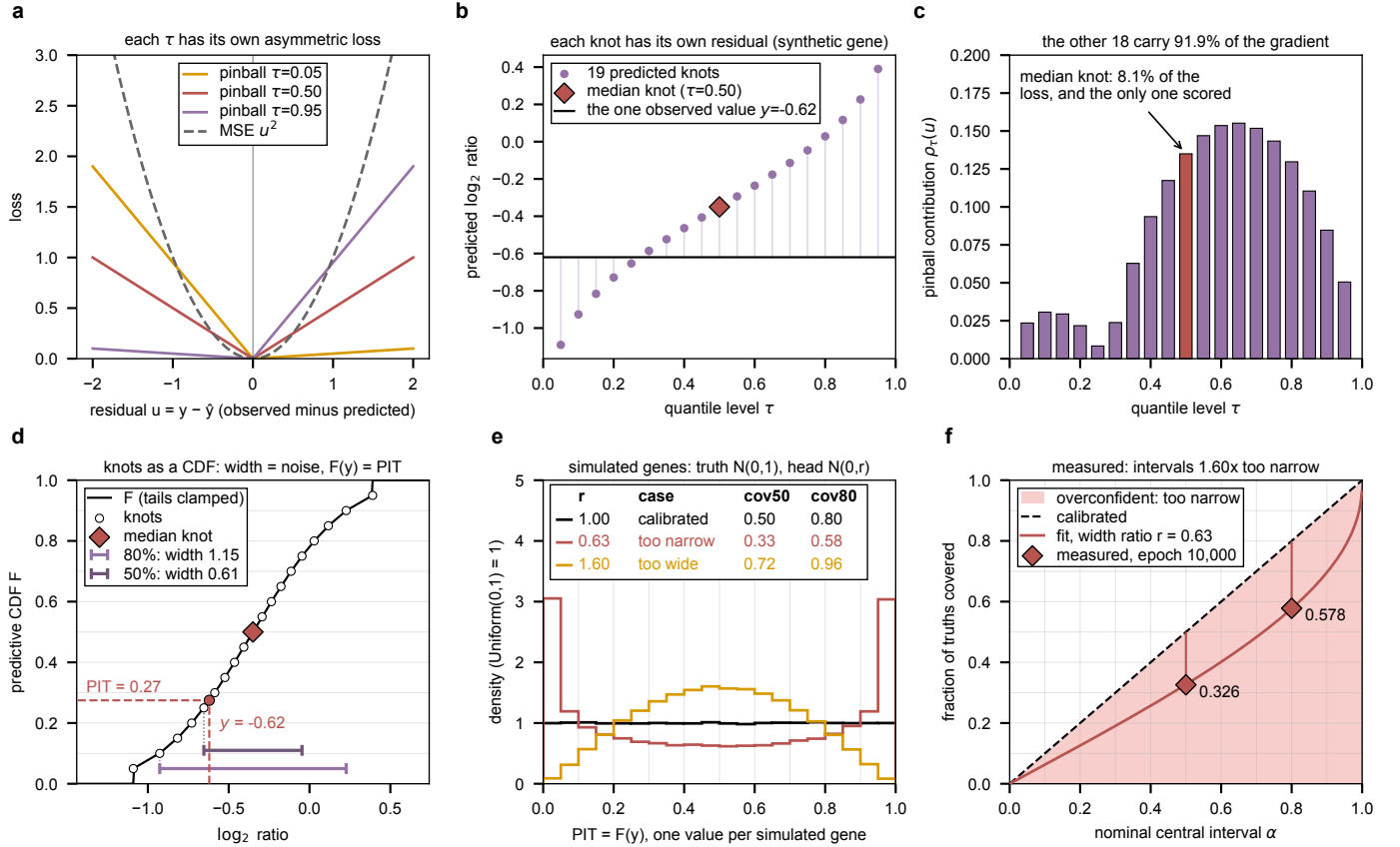
| epochs | mean   | sd     | main effect, 32 v 32 | cell, 16 v 16 | arm, 8 v 8 |
|--------|--------|--------|----------------------|---------------|------------|
| 350    | 0.1198 | 0.0096 | 0.0067               | 0.0095        | 0.0135     |
| 500    | 0.1223 | 0.0058 |                      |               |            |
| 700    | 0.1411 | 0.0166 |                      |               |            |
| 1,000  | 0.1609 | 0.0099 |                      |               |            |
| 1,500  | 0.1670 | 0.0117 |                      |               |            |
| 2,000  | 0.1745 | 0.0151 |                      |               |            |
| 2,800  | 0.1821 | 0.0175 |                      |               |            |
| 4,000  | 0.1883 | 0.0171 |                      |               |            |
| 6,000  | 0.1917 | 0.0177 |                      |               |            |
| 9,900  | 0.1965 | 0.0222 |                      |               |            |

**Table 10:** Mean and spread of the eight long-budget v9 arms at truncated budgets, from 500-sample curves. The spread is not monotone in budget: 700 epochs is the noisiest point below 2,000. The last three columns are the detectable gap at the 350-epoch spread for the three contrast sizes a  $2^4$  grid affords, the same arithmetic that sized the v10 grid at 1,400 epochs and two seeds rather than 700 and four.

Two traps were hit measuring this and are recorded so the next reader does not hit them: a resumed run passes an epoch filter and inherits its parent’s config, so it masquerades as a replicate and is identified only by where its curve starts; and a whole-curve-maximum collapse test reads a run that collapses late as healthy, so collapse is tested on the last value.

## 2.5 A short screen does not rank like a long run, within one configuration x

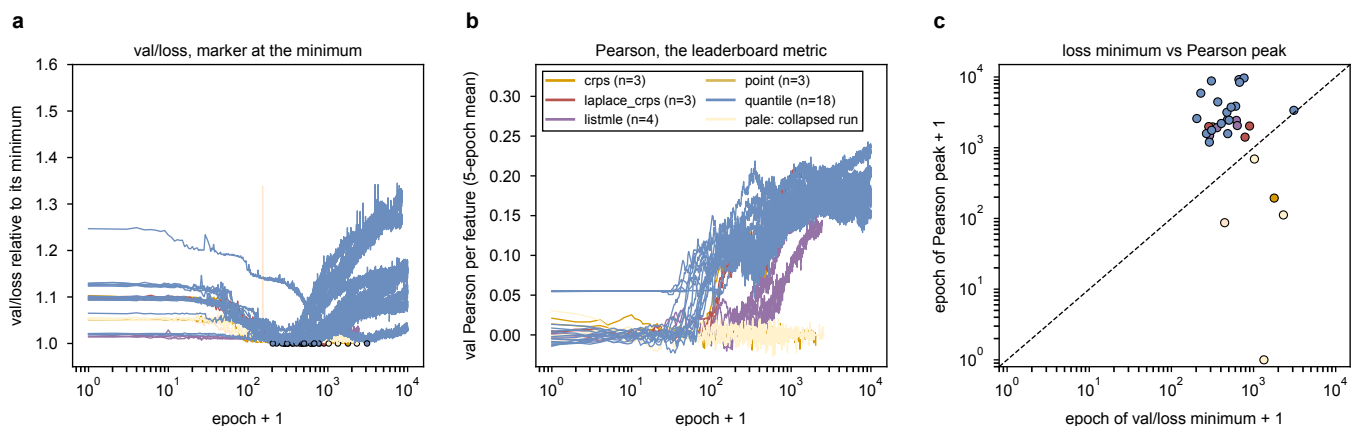
Spearman between the 700-epoch and 3,000-epoch `roll_max` of the same run (`budget_rank_preservation.py`):  $-0.139$ ,  $p = 0.53$ ,  $n = 23$ ; top-3 overlap one of three, top-5 one of five. After resume exclusion the corpus is the eight mask-schedule arms plus the objective round, so this measures persistence among near-identical configurations: a run’s early rank among them says nothing about its late rank. Whether a short screen orders different configurations correctly is answerable only on a multi-config corpus, and the v10 grid (section 3.1) is the first one.



**Figure 1: The metric reads one knot of nineteen, and the eighteen it ignores set intervals the incumbent makes 1.6 times too narrow.** Panels **a** to **e** are synthetic; **f** is measured. **a**, the pinball loss at three quantile levels against squared error. **b**, one illustrative gene: 19 predicted knots, the median knot, and the one observed value; each knot's residual to that value feeds its own pinball term. **c**, the 19 terms; the median carries 8.1% and is the only knot the metric scores. **d**, the same knots read back as a CDF (joined piecewise-linearly, tails clamped as `_quantile_pit` does): the central 80% and 50% interval widths are the model's own noise estimate for that gene, and the CDF at the observed value is its PIT. **e**, PIT histograms over 200,000 simulated genes with truth  $N(0,1)$  and a head predicting  $N(0,r)$ : flat when  $r = 1$ , U-shaped when the intervals are too narrow, humped when too wide, with the coverage of the central 50% and 80% intervals per case. **f**, the incumbent's measured coverage at epoch 10,000 as a reliability curve; the fitted curve is a single width ratio  $r = 0.63$ , derived not measured. Generated by `plot_pinball_vs_mse.py`.

## 2.6 The validation loss bottoms out early and the metric keeps rising ✗

Full per-epoch history of every live v9 run whose loss is not the metric, 27 runs of 1,000 epochs or more that neither collapsed nor resumed, read 2026-09-09 (`loss_min_vs_pearson_peak.py`, fig. 2); the four `listmle` runs that joined the objective round after table 9 was read are included, and both `listmle_mse` runs collapsed. The median epoch of the `val/loss` minimum is 481 (range 203 to 3,097) and no run bottoms out by epoch 100. The median final loss sits 9.0% above its minimum. Every run's Pearson peak (median epoch 2,433) comes after its loss minimum, at a loss already 6.8% above the floor, and the median Pearson gained between the two is +0.057. The long budget buys Pearson on a rising validation objective, which is the fact the metric-aligned round was built to confront. The eight Pearson-objective runs are excluded here by construction, since their loss and metric coincide on train.



**Figure 2: The validation loss bottoms out before the Pearson peaks in every run, so the long budget buys Pearson on a rising objective.** **a**, validation loss relative to its minimum, one curve per run, colored by head; pale curves are collapsed runs. **b**, the 5-epoch rolling mean of validation Pearson. **c**, the epoch of the loss minimum against the epoch of the Pearson peak; every point lies above the diagonal. Generated by `loss_min_vs_pearson_peak.py`.

## 2.7 Every run behind these numbers ✗

Table 11 links every run this document draws a number from to its W&B page, generated by `wandb_run_index.py` from the same result files the sections read.

## 2.8 Training on the metric does not resolve it ✗

Two heads were added to `torchcell/losses/distributional.py` on 2026-09-06: `pearson`, whose loss is one minus the mean per-feature Pearson over the genes hidden at the current unmasking step, each gene's correlation taken over the strains in the batch where it is hidden, with columns of fewer than three scored rows or constant prediction or target dropped as the metric drops them; and `pearson_mse`, the same plus the masked squared error at unit weight as a scale anchor. Both are point-shaped, carry no PIT and no calibration keys, and the pure loss is scale-free, so its `mse`, `nmse` and spread ratio are not comparable to any other arm's. Eighty-six unit tests and a container canary (2375312) preceded the launch. The readout is section 3.3: collapse at every batch-32 seed of the pure loss, at two of three seeds of the anchored one, and parity with the incumbent for the batch-64 survivors, which peak by epoch 2,400 and give part of it back.

A third pair followed on 2026-09-08: `listmle`, the Plackett-Luce negative log-likelihood of the true ordering of the strains in the batch, per hidden gene, divided by the list length (Xia et al. 2008), and `listmle_mse`, the same plus the masked squared error at unit weight. The question it asks is the ranking one: given a set of knockouts, which strain expresses gene  $g$  highest. ListMLE keeps a gradient on a constant column ( $\log(n!)/n$  per gene, where Pearson drops the column), is shift-invariant but not scale-invariant, so predicted spread can only grow under it, and makes the batch part of the objective as Pearson does. Ninety-eight unit tests and a canary (2385788) preceded the launch; the readout so far is section 3.4.

**Table 11:** Every run behind the numbers in this document, 80 in all, in projects [torchcell\\_019\\_expr\\_v9](#) and [torchcell\\_019\\_expr\\_v10](#) (entity [zhao-group](#)). One row per round and arm; each entry is a seed and its run id, linked to the run page, with the final epoch in brackets. The objective round lists each fresh run and its resume. The v10 grid names cells by embedding, trunk, readout and weight decay.

| Round          | Arm                    | Seed: run [epochs]                                                                                                                                                                                                      |
|----------------|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v9 mask arms   | M_coarse               | 0: <a href="#">ebkzn1ao</a> [9,900]                                                                                                                                                                                     |
|                | M_fine                 | 0: <a href="#">hx8pxdic</a> [9,900]                                                                                                                                                                                     |
|                | M_gate_rezero          | 0: <a href="#">u1vuznme</a> [9,900]                                                                                                                                                                                     |
|                | M_hi                   | 0: <a href="#">da5g4o9v</a> [9,900]                                                                                                                                                                                     |
|                | M_lo                   | 0: <a href="#">f2wf23oy</a> [9,900]                                                                                                                                                                                     |
|                | M_nomix                | 0: <a href="#">rb3bhryq</a> [9,900]                                                                                                                                                                                     |
|                | M_off                  | 0: <a href="#">tow1z48n</a> [9,900]                                                                                                                                                                                     |
|                | M_sched                | 0: <a href="#">8r5ewoaq</a> [9,900]                                                                                                                                                                                     |
| objective      | Q_point                | 0: <a href="#">8gklzy7n</a> [9,895] 0: <a href="#">iiqohck6</a> [2,039] 1: <a href="#">yrs7x07y</a> [9,895] 1: <a href="#">qplil73c</a> [2,039] 2: <a href="#">y8ex71vn</a> [9,863] 2: <a href="#">he2vb4qr</a> [2,577] |
|                | Q_crps                 | 0: <a href="#">m6utaeu8</a> [9,896] 0: <a href="#">1ljkhck</a> [1,995] 1: <a href="#">djgkdzol</a> [9,895] 1: <a href="#">nx5dwhp8</a> [2,039] 2: <a href="#">eiu664kx</a> [9,888] 2: <a href="#">hzumvs8y</a> [2,031]  |
|                | Q_laplace              | 0: <a href="#">1xa3x8uc</a> [9,896] 0: <a href="#">c981k5n1</a> [1,995] 1: <a href="#">7v916ett</a> [9,895] 1: <a href="#">ia6312dv</a> [2,039] 2: <a href="#">g69wskm0</a> [9,888] 2: <a href="#">1z8gq5ui</a> [2,031] |
| mechanism      | R_ref                  | 0: <a href="#">i63dbvha</a> [4,139] 1: <a href="#">r2s57bvt</a> [8,499]                                                                                                                                                 |
|                | R_basis64              | 0: <a href="#">h9l8aguy</a> [8,499] 1: <a href="#">4gtnrja7</a> [7,018]                                                                                                                                                 |
|                | R_pergene              | 0: <a href="#">4u0txt3s</a> [8,499] 1: <a href="#">vqek7ali</a> [8,499]                                                                                                                                                 |
|                | R_pergene_basis64      | 0: <a href="#">nrkorko0</a> [4,079] 1: <a href="#">3qy1rh0o</a> [6,589]                                                                                                                                                 |
| metric-aligned | Q_pearson              | 0: <a href="#">zaul43n1</a> [4,298] 1: <a href="#">kspeoljg</a> [4,291] 2: <a href="#">p2qq2525</a> [3,166]                                                                                                             |
|                | Q_pearson_mse          | 0: <a href="#">hsauh7al</a> [4,298] 1: <a href="#">ppc2pyv5</a> [6,576] 2: <a href="#">4ei42tb1</a> [3,169]                                                                                                             |
|                | Q_pearson_b64          | 0: <a href="#">7ylecrjz</a> [9,118] 1: <a href="#">wb2xocf2</a> [9,114]                                                                                                                                                 |
| ranking        | Q_listmle              | 0: <a href="#">teaeym0k</a> [1,959] 1: <a href="#">g755dzhn</a> [1,792]                                                                                                                                                 |
|                | Q_listmle_mse          | 0: <a href="#">dhqpktod</a> [637] 1: <a href="#">j6m31tov</a> [480]                                                                                                                                                     |
|                | Q_listmle_b64          | 0: <a href="#">k79hvcqe</a> [2,438] 1: <a href="#">6b33ftkt</a> [2,437]                                                                                                                                                 |
| v10 grid       | c0_rand_big_mlp_wd0    | 0: <a href="#">dd26f5nw</a> [1,399] 1: <a href="#">nxrpgwca</a> [1,176]                                                                                                                                                 |
|                | c1_ptt5_big_mlp_wd0    | 0: <a href="#">te8272kk</a> [1,399] 1: <a href="#">14uw0g1e</a> [1,399]                                                                                                                                                 |
|                | c2_rand_small_mlp_wd0  | 0: <a href="#">j5oc0kbf</a> [1,399] 1: <a href="#">zu0mp0ly</a> [1,399]                                                                                                                                                 |
|                | c3_ptt5_small_mlp_wd0  | 0: <a href="#">lnoei9xt</a> [1,399] 1: <a href="#">wv0qlp4n</a> [1,192]                                                                                                                                                 |
|                | c4_rand_big_lin_wd0    | 0: <a href="#">0htrkihg</a> [1,119] 1: <a href="#">59a4xytq</a> [1,399]                                                                                                                                                 |
|                | c5_ptt5_big_lin_wd0    | 0: <a href="#">6ee07w1j</a> [1,399] 1: <a href="#">7ue6aduu</a> [990]                                                                                                                                                   |
|                | c6_rand_small_lin_wd0  | 0: <a href="#">6ktjsou6</a> [1,399] 1: <a href="#">ujbunglx</a> [1,020]                                                                                                                                                 |
|                | c7_ptt5_small_lin_wd0  | 0: <a href="#">dcx819z6</a> [1,148] 1: <a href="#">ts735yhp</a> [1,399]                                                                                                                                                 |
|                | c8_rand_big_mlp_wd2    | 0: <a href="#">fk6luj3g</a> [1,119] 1: <a href="#">7n0ohmht</a> [1,399]                                                                                                                                                 |
|                | c9_ptt5_big_mlp_wd2    | 0: <a href="#">5ivl9nat</a> [1,399] 1: <a href="#">iyvyd7c3</a> [990]                                                                                                                                                   |
|                | c10_rand_small_mlp_wd2 | 0: <a href="#">p2dj91nf</a> [1,399] 1: <a href="#">yecoyfgc</a> [1,029]                                                                                                                                                 |
|                | c11_ptt5_small_mlp_wd2 | 0: <a href="#">oe77kyf6</a> [1,148] 1: <a href="#">hhvof18w</a> [1,399]                                                                                                                                                 |
|                | c12_rand_big_lin_wd2   | 0: <a href="#">kt9tezrk</a> [1,399] 1: <a href="#">rd64e6kx</a> [1,170]                                                                                                                                                 |
|                | c13_ptt5_big_lin_wd2   | 0: <a href="#">r9w7i1fw</a> [1,399] 1: <a href="#">c3xrtsr7</a> [1,399]                                                                                                                                                 |
|                | c14_rand_small_lin_wd2 | 0: <a href="#">jrwpcoxd</a> [1,399] 1: <a href="#">4ity6oib</a> [1,399]                                                                                                                                                 |
|                | c15_ptt5_small_lin_wd2 | 0: <a href="#">a6mmvj8i</a> [1,399] 1: <a href="#">04204r3z</a> [1,198]                                                                                                                                                 |

### 3 The readouts: four rounds, one large effect ✗

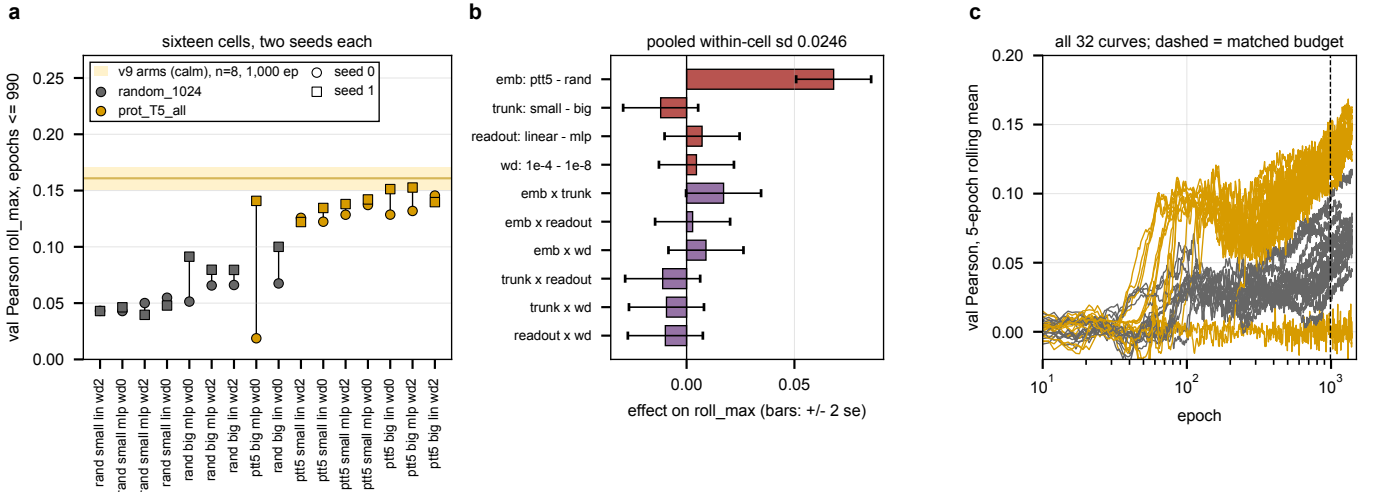
Three rounds returned within two days of each other on 2026-09-08, the  $2^4$  factorial on Delta, the mechanism round on IGB (both sized by the arithmetic of section 1.6) and the metric-aligned objective round, which trains on  $1 - r$  directly (section 2.8); the ranking round that replaced the dead half of the metric-aligned one is a day old at this read. Every score below is `roll_max`, the maximum of a centered 5-epoch rolling mean of `val/expression/pearson_per_feature`, read from the full per-epoch history, and every contrast is at a MATCHED budget computed as the smallest final epoch across the runs being compared. The scoring rule is an upward-biased order statistic; it is the same rule on every side of every contrast, so the bias cancels in a difference and not in an absolute number. Runs still training are read at the epoch they had reached, stated with each number, and W&B’s `finished` state is not evidence that a run ended: these runs train offline on IGB and every sync from the login node stamps the snapshot `finished`. Whether a job is alive is read from `squeue`.

#### 3.1 The factorial: embedding content is the only factor that moves the score ✗

Job 21830323 ran 16 cells at two seeds, one run per GPU, four per node. Five of the eight array tasks completed 1,400 epochs; three hit the two-day wall between epochs 990 and 1,198, so the matched budget is epochs  $\leq 990$ .

| factor                    | contrast (level 1 – level 0)                        | effect | $t$  | effect, 31 runs | $t$   |
|---------------------------|-----------------------------------------------------|--------|------|-----------------|-------|
| embedding                 | <code>prot_T5_all</code> – <code>random_1024</code> | +0.068 | +7.8 | +0.076          | +17.3 |
| trunk                     | $L=2, h=45$ – $L=6, h=90$                           | –0.012 | –1.4 | –0.018          | –4.0  |
| readout (inert, see text) | linear – two-layer                                  | +0.007 | +0.8 | +0.002          | +0.5  |
| weight decay              | $10^{-4}$ – $10^{-8}$                               | +0.005 | +0.5 | –0.000          | –0.1  |
| embedding $\times$ trunk  |                                                     | +0.017 | +2.0 | +0.013          | +2.9  |

**Table 12:** Main effects of the v10 grid at epochs  $\leq 990$ , 16 runs a side. The error is the pooled within-cell replicate standard deviation, 0.0246 on 16 degrees of freedom over all 32 runs (standard error per effect 0.0087), and 0.0122 on 15 degrees of freedom once the one run that never left the chance band is dropped (standard error 0.0044). The other five two-way interactions have  $|t| < 1.3$  in both readings.



**Figure 3: Embedding content is the one factor that moves the score at a matched budget (+0.07, three times the replicate spread); trunk size, readout and weight decay sit within it (b). a**, every cell’s two seeds at epochs  $\leq 990$ , sorted by cell mean and colored by embedding, against the eight long-budget v9 arms at 1,000 epochs ( $0.161 \pm 0.010$ , calm embeddings, band is one standard deviation across arms). **b**, main effects and two-way interactions with  $\pm 2$  standard errors from the pooled within-cell spread. **c**, all 32 validation curves; the dashed line is the matched budget. Generated by `v10_grid_factorial.py`.

**Embedding content is worth 0.07, three times the replicate spread.** The ProtT5 cells average 0.129 against 0.061 for random vectors of the same width, and no cell of one level overlaps any cell of the other (fig. 3a). The contrast was built as a content test at matched width 1,024, so it says the content matters and says nothing about calm, the incumbent’s embedding, which is not a level of the grid. The one healthy draw of the incumbent-with-ProtT5 cell scores 0.141 against the incumbent’s  $0.161 \pm 0.010$  at the same budget, one draw against eight, and that is not a resolved gap in either direction.

**Weight decay is a measured null; the readout row is retracted.** The weight-decay effect is under 0.008 against a resolution of about 0.02 at two standard errors. The readout row is a null by construction, not a measurement: `linear_readout` was declared in the config and carried in every run’s recorded configuration, but the model never passed it to the head, so both levels trained the two-layer head. The parameter counts logged to W&B are identical across the two levels, 1,447,309 under the deep trunk and 682,988 under the shallow one, which is the tell. The option was wired on 2026-09-10 with a regression test, and linear against two-layer readout is unmeasured until the v12 readout round (section 5). The trunk is a small effect in favor of the deeper one,  $-0.012$  over all runs and  $-0.018$  once the stuck run is dropped, and it is sharper under ProtT5 (the interaction). Of the three factors that were in fact varied, one acts.

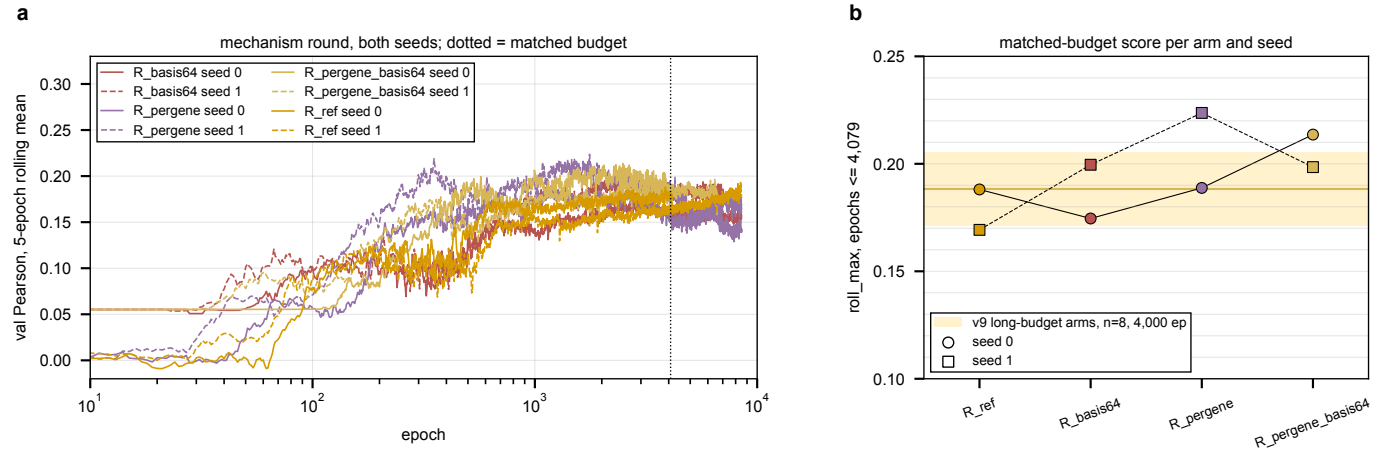
**One run in 32 never learned.** Cell 1 seed 0 (`te8272kk`: ProtT5,  $L=6$ , two-layer readout, weight decay  $10^{-8}$ ) sits at 0.019 at its best and  $-0.007$  at epoch 1,399 with `nmse` 1.010, while its seed-1 twin is the grid’s best single run at 0.169. That pair alone carries most of the pooled variance, which is why both readings are given. Its cause is not measured.

### 3.2 The mechanism round: per-gene readout $+0.03$ , under the round’s resolution $\times$

All four array tasks of the round ended in the cgroup out-of-memory handler at host RSS 61 GB against a 60 GB request, after 3.5 to 4.6 days. In each task one of the two packed runs was killed and the other finished its 8,500 epochs, so the matched budget is set by the shortest survivor, epochs  $\leq 4,079$ . The two seed-1 runs abandoned on cabbi at epoch  $\approx 1,595$  when that half moved to the A40 node are superseded, not replicates, and are dropped.

| arm                            | seed 0 | seed 1 | paired difference vs <code>R_ref</code> (s0 / s1) | mean     |
|--------------------------------|--------|--------|---------------------------------------------------|----------|
| <code>R_ref</code>             | 0.188  | 0.169  |                                                   |          |
| <code>R_basis64</code>         | 0.175  | 0.200  | $-0.014$ / $+0.030$                               | $+0.008$ |
| <code>R_pergene</code>         | 0.189  | 0.224  | $+0.001$ / $+0.054$                               | $+0.028$ |
| <code>R_pergene_basis64</code> | 0.214  | 0.199  | $+0.026$ / $+0.029$                               | $+0.027$ |

**Table 13:** The mechanism round at epochs  $\leq 4,079$ , paired within seed against the in-round reference. Seed 0 ran on cabbi Ada cards, seed 1 on one A40 node, so GPU type is a seed-level block. The eight long-budget v9 arms score  $0.188 \pm 0.017$  at 4,000 epochs and both `R_ref` draws fall inside that band.



**Figure 4:** No mechanism arm separates from the reference at two seeds; the largest paired gain,  $+0.03$ , is the size of the seed-to-seed spread. **a**, validation Pearson of all eight runs (solid seed 0, dashed seed 1); the dotted line is the matched budget. **b**, the matched-budget score per arm and seed against the long-budget arm band at 4,000 epochs. Generated by `mech_round_readout.py`.

**The pair term alone is a null.** Its two paired differences disagree in sign and average  $+0.008$ . The per-gene readout arms average  $+0.027$ , and the combined arm is positive at both seeds, but the round was sized to resolve about 0.06 at two pairs (table 10), so this is a direction and not a result. What every curve does show is the dynamics: the per-gene arms reach their peak between epochs 1,200 and 2,600 and give part of it back (`R_pergene` ends at 0.137 and 0.170 against peaks of 0.189 and 0.224), while the reference and basis arms are flat or still rising at 8,500. Whether the give-back is overfitting of the extra per-gene parameters is a hypothesis and is not measured.

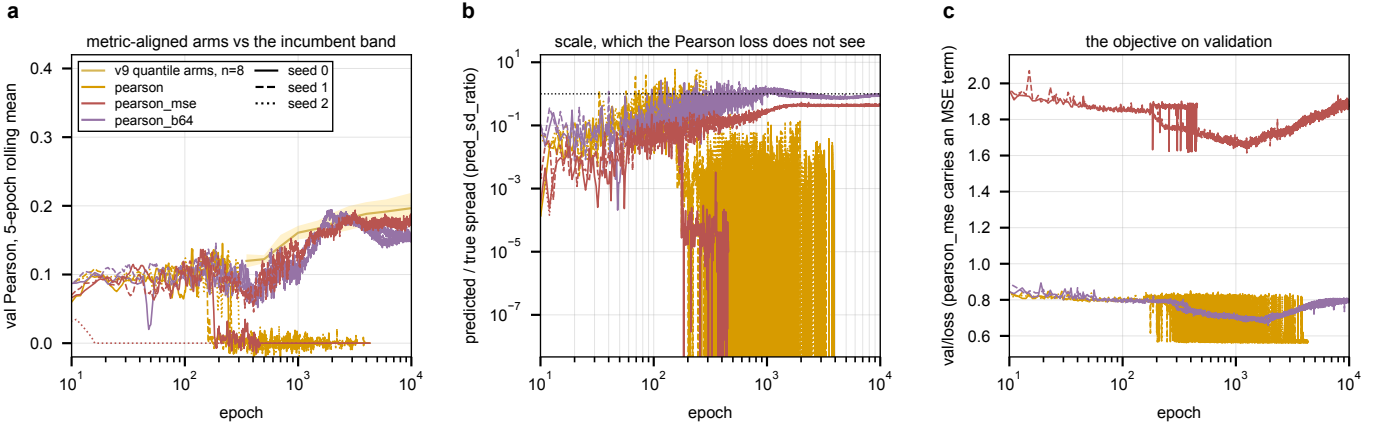
**A tagging defect, found in the readout.** The arm script carried a wave-4b `R_ref` branch ahead of the wave-5 one; a shell `case` takes its first match, so every reference run of this round is tagged `stage-wave4b`. The readout selects by arm and config tag, and the shadowing branch is removed.

### 3.3 The metric-aligned round at day three: collapse is the rule at batch 32 ✗

Three arms train on  $1 - r$  directly, the mean per-feature Pearson over the genes hidden at the current unmasking step (**pearson**), the same plus a masked squared error at unit weight (**pearson\_mse**), and the pure loss at batch 64 (**pearson\_b64**), on IGB cabbi from 2026-09-06. The two batch-32 arms run three seeds at two runs per card, the batch-64 arm two seeds solo. Five of the six batch-32 runs were on the floor by day two and were stopped on 2026-09-08 (two whole tasks canceled, one dead process killed inside its task so its live sibling kept the card); at the read of 2026-09-09 18:15 CT the three survivors are still training, the batch-64 pair at 9,114 and 9,118 of 9,900 and the anchored seed-1 run at 6,576. The readout script records the epoch each reached, and every number below is partial at that epoch.

| arm         | seed | epoch  | roll_max @ epoch | Spearman | last  | on the floor from | spread ratio |
|-------------|------|--------|------------------|----------|-------|-------------------|--------------|
| pearson     | 0    | 4,298  | 0.145 @ 215      | 0.092    | 0.000 | 584               | 0.000        |
| pearson     | 1    | 4,291  | 0.133 @ 151      | 0.085    | 0.000 | 2,203             | 0.000        |
| pearson     | 2    | 3,166  | 0.137 @ 242      | 0.086    | 0.000 | 680               | 0.000        |
| pearson_mse | 0    | 4,298  | 0.137 @ 138      | 0.086    | 0.000 | 360               | —            |
| pearson_mse | 1    | 6,576* | 0.194 @ 3,193    | 0.178    | 0.172 | never             | 0.43         |
| pearson_mse | 2    | 3,169  | 0.035 @ 5        | 0.030    | 0.000 | 14                | —            |
| pearson_b64 | 0    | 9,118* | 0.194 @ 1,926    | 0.175    | 0.148 | never             | 0.89         |
| pearson_b64 | 1    | 9,114* | 0.186 @ 2,419    | 0.171    | 0.160 | never             | 0.90         |

**Table 14:** The metric-aligned round at day three; \* marks a run still training at the read. Spearman is the **roll\_max** of the per-feature Spearman, the quantity the ranking round targets. “On the floor from” is the first epoch after which the 5-epoch rolling mean of the validation Pearson stays under 0.02; the seed-1 **pearson** run is near zero from epoch  $\approx 250$  and flickers above the line until 2,203. The spread ratio is predicted over true standard deviation (**pred\_sd\_ratio**) at the last epoch. The eight long-budget v9 arms score  $0.188 \pm 0.017$  at 4,000 epochs,  $0.192 \pm 0.018$  at 6,000 and  $0.1965 \pm 0.0222$  at 9,900.



**Figure 5: Training on the Pearson objective does not beat the pinball incumbent: one run sits at the floor from epoch 250 and the rest stay inside the long-budget band.** **a**, validation Pearson of all eight runs against the long-budget arm band. **b**, predicted over true spread, which the Pearson loss cannot see; a run whose outputs go constant falls off the bottom. **c**, the objective on validation; **pearson\_mse** carries the squared-error term and sits on its own scale. Generated by `pearson_round_readout.py`.

**Five of six batch-32 runs are on the floor, and the loss does not see it.** The three pure-Pearson runs peak between 0.13 and 0.145 at epochs 150 to 240 and then fall, their predicted spread dropping from about 0.3 to  $10^{-7}$  over the same epochs (fig. 5b): the network’s output goes constant per gene. Their validation loss holds near 0.56 throughout, because the loss drops a constant column as invalid and averages over the columns still varying, so it reads  $1 - 0.44$  on a handful of genes while the metric reads 0 on all of them. The failure mode the design named is the one that occurred, with the difference that the loss never reads its connected zero. The squared-error anchor at unit weight did not prevent it: **pearson\_mse** collapsed by epoch 14 at one seed and 360 at another, and its surviving seed reads **roll\_max** 0.194 at epoch 6,576 with a last value of 0.172, inside the long-budget arm band and still training.

**The batch-64 runs survive and sit on the band, not above it.** At 9,114 and 9,118 epochs, with spread ratios of 0.89 and 0.90, they read **roll\_max** 0.194 and 0.186 against the incumbent’s  $0.1965 \pm 0.0222$  at 9,900, peaked near epochs 1,900 to 2,400 and sit at 0.148 and 0.160 now, 0.03 to 0.05 below their peaks with under 800 epochs to run. Batch size and solo placement are confounded in that arm, so why it survives is not measured; the in-batch correlation being taken over twice as many strains is a hypothesis. Nothing in the round beats the quantile head at any budget read so far. The question it was launched to answer, whether the metric’s late rise is generalization gap or objective

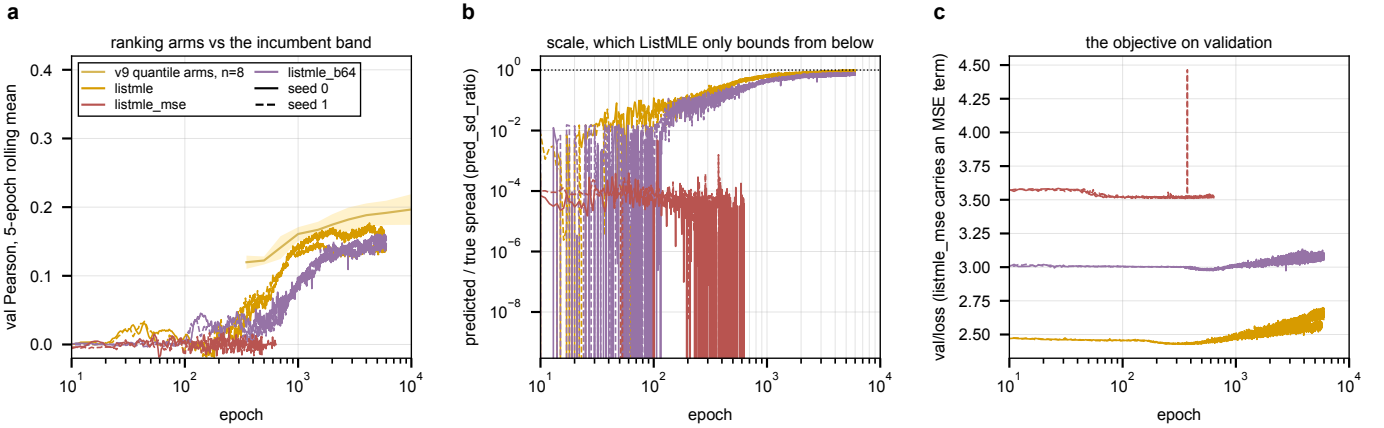
disagreement, is not answered by two survivors that peaked by epoch 2,400: on the Pearson objective the late rise did not happen, and whether that is the objective or the batch size is the confound above.

### 3.4 The ranking round at day one: ListMLE learns, slowly, and does not collapse ✗

Three arms replaced the dead half of the metric-aligned round on 2026-09-08 20:05 CT: `listmle` at batch 32, two seeds packed on cabbi with the anchored `listmle_mse` beside them, and `listmle` at batch 64 solo on two A40s, all at a 6,000-epoch budget chosen because every packed five-day task so far died of host memory between days 3.5 and 4.6 and the Pearson peaks had come by 3,200. Read at 2026-09-09 18:15 CT with every surviving run still training (`pearson_round_readout.py--roundlistmle`).

| arm                      | seed | epoch  | roll_max @ epoch | Spearman | last   | on the floor from | spread ratio |
|--------------------------|------|--------|------------------|----------|--------|-------------------|--------------|
| <code>listmle</code>     | 0    | 1,959* | 0.168 @ 1,920    | 0.173    | 0.167  | never             | 0.80         |
| <code>listmle</code>     | 1    | 1,792* | 0.150 @ 1,488    | 0.154    | 0.140  | never             | 0.75         |
| <code>listmle_b64</code> | 0    | 2,438* | 0.144 @ 2,433    | 0.134    | 0.137  | never             | 0.67         |
| <code>listmle_b64</code> | 1    | 2,437* | 0.150 @ 2,057    | 0.138    | 0.144  | never             | 0.67         |
| <code>listmle_mse</code> | 0    | 637    | 0.025 @ 1        | 0.024    | 0.000  | 3                 | 0.000        |
| <code>listmle_mse</code> | 1    | 480    | 0.024 @ 246      | 0.024    | -0.033 | 247               | 0.000        |

**Table 15:** The ranking round at day one; \* marks a run still training. Columns as in table 14. The eight long-budget v9 arms score  $0.1745 \pm 0.0151$  at 2,000 epochs and  $0.1821 \pm 0.0175$  at 2,800. The two anchored predictor runs sat at the mean predictor from epoch 160 and were killed by process at 03:40 CT on 2026-09-09 so their pure siblings could take the whole card.



**Figure 6:** The anchored ranking runs collapse to the mean predictor by epoch 250 and the pure ranking runs stay under the long-budget band. **a**, validation Pearson of all six runs against the long-budget arm band. **b**, predicted over true spread; the anchored runs fall off the bottom by epoch 250 while the pure runs climb from 0.05 toward 1. **c**, the objective on validation, which bottoms at epochs 290 to 630 and rises after, as every other objective in fig. 2 does. Generated by `pearson_round_readout.py--roundlistmle`.

**The anchor kills it, the pure loss does not.** With the squared error at unit weight both seeds went to the per-gene mean by epoch 160 and stayed there; the MSE gradient is about  $50\times$  the ranking gradient at initialization and reaches its own optimum, the constant, which the ranking term’s  $\log(n!)/n$  floor does not pull it off. That is the mirror image of the Pearson round, where the anchor was what kept the one surviving batch-32 run alive. The pure ranking loss has not collapsed at any of its four runs, and its spread ratio climbs, from under 0.05 before epoch 200 to 0.80 at 1,959, as a shift-invariant objective that rewards larger score gaps should make it; the leaderboard’s `nmse` of 74.7 for these runs (table 9) is that free location, not a failure to order.

**Behind the band at matched epochs, by 0.01 to 0.04.** At epochs  $\leq 2,000$  the batch-32 pair reads 0.168 and 0.150 against the long-budget arms’  $0.1745 \pm 0.0151$ , one seed inside the band and one below it; the batch-64 pair at 2,400 reads 0.144 and 0.150 against  $0.1821 \pm 0.0175$  at 2,800, both below. Every curve is still rising and none has peaked, so these are epochs reached and not results. On the quantity ListMLE targets, the per-feature Spearman, the batch-32 seed-0 run reads 0.173 at 1,920 epochs where the surviving Pearson-objective runs read 0.171 to 0.178 at 6,500 to 9,100; whether that is a real ordering advantage or a matched-epoch artifact is answerable when the round ends. Hypothesis (untested): the ranking gradient is small in absolute terms, norm 0.3 at initialization, and the learning rate pinned at  $3 \times 10^{-4}$  for the pinball loss is low for it, which would show as the slow start of fig. 6a.

### 3.5 The readout round at day two: the State SE form leads by 0.01 in every seed, nothing is resolved ✗

The round designed in section 5 (eight per-gene readouts on the v11 E\_full embedding stack under the pinball objective, four seeds, four runs per card) ran on IGB from 2026-09-10 23:17 CT. The six A40 tasks carrying seeds 1 to 3 finished between 09:54 and 11:02 CT on 2026-09-12 after 1 d 10 h to 1 d 12 h, every run at its 1,400 epochs (about 40 epochs per hour per run at four per card, 2.6 times the solo rate in throughput); the two cabbi tasks carrying seed 0 (RTX 6000 cards, so GPU type is a seed-level block as in the mechanism round) finished at 16:19 and 16:44 CT the same day after 1 d 17 h. All 32 runs were synced from the login node and read out with the round's rule: `roll_max` at the matched budget of 1,399 epochs, paired within seed against `H_ref`, the shared two-layer MLP readout.

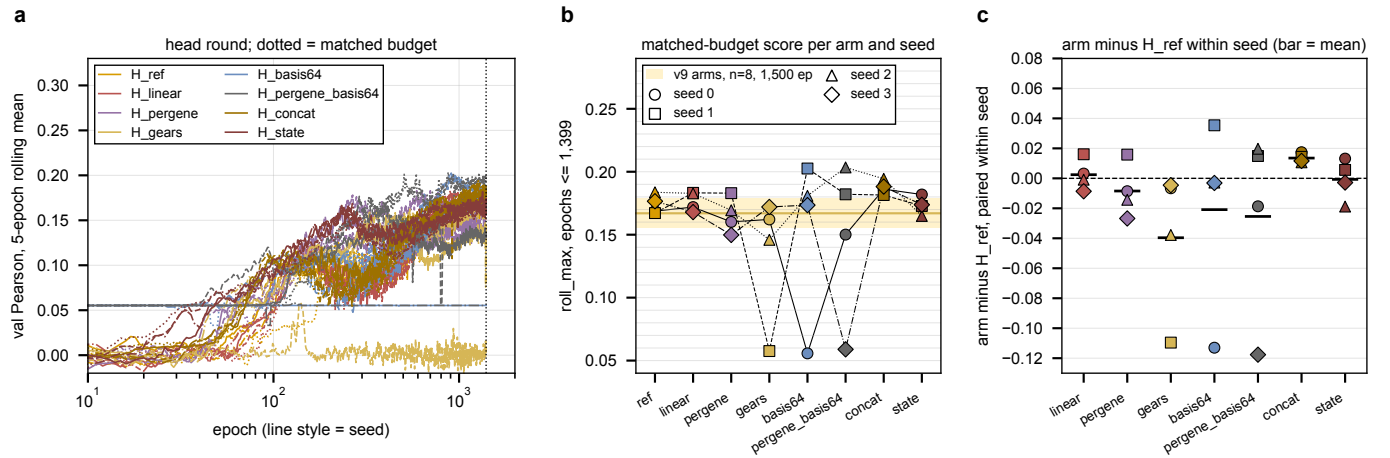
SRC: `experiments/019-simb-multimodal/results/head_round_readout.json`

**Table 16:** The head round at epochs  $\leq 1,399$ , four seeds. `roll_max` per arm (mean and sd over four seeds) and the paired difference against `H_ref` within seed; a star marks a run that never left the constant plateau. The v9 long-budget band at the nearest tabulated budget (1,500 epochs) is  $0.1670 \pm 0.0117$  over eight arms.

| arm                                     | mean   | sd     | minus <code>H_ref</code> , seeds 0 / 1 / 2 / 3 | mean diff |
|-----------------------------------------|--------|--------|------------------------------------------------|-----------|
| <code>H_ref</code> (shared MLP)         | 0.1740 | 0.0076 |                                                |           |
| <code>H_linear</code>                   | 0.1765 | 0.0077 | +0.003 / +0.016 / -0.001 / -0.009              | +0.002    |
| <code>H_pergene</code> (GEARS row)      | 0.1656 | 0.0141 | -0.009 / +0.016 / -0.014 / -0.027              | -0.008    |
| <code>H_gears</code> (row + cross-gene) | 0.1344 | 0.0523 | -0.007 / -0.110* / -0.038 / -0.005             | -0.040    |
| <code>H_basis64</code> (scGPT MVC form) | 0.1532 | 0.0661 | -0.113* / +0.035 / -0.003 / -0.003             | -0.021    |
| <code>H_pergene_basis64</code>          | 0.1486 | 0.0637 | -0.019 / +0.015 / +0.020 / -0.118*             | -0.025    |
| <code>H_concat</code> (State SE form)   | 0.1876 | 0.0053 | +0.017 / +0.014 / +0.011 / +0.012              | +0.0135   |
| <code>H_state</code> (State ST form)    | 0.1733 | 0.0070 | +0.013 / +0.006 / -0.019 / -0.003              | -0.001    |

**Three runs never trained.** `H_gears` seed 1 collapsed to constant output (validation Pearson 0.000, predicted-to-true spread ratio 0.000 at the end), `H_pergene_basis64` seed 3 sat on the 0.059 mean plateau for all 1,399 epochs, and `H_basis64` seed 0 did the same at 0.056; the three were the runs flagged as stuck at 14.7 h. They are the starred entries in table 16, and they are what carries those three arms' means down; the escape-from-plateau failure rate at four runs per card, three of 32, is itself a number the round returns, and all three are basis or cross-gene forms.

**What the twenty-nine that trained say.** `H_concat`, the State SE decoder form (the strain vector concatenated to every gene token before the readout), is the only arm above `H_ref` in every seed, by +0.011 to +0.017, mean +0.0135 with a seed-to-seed sd of 0.0026 on the difference, a paired  $t$  of about 10 on three degrees of freedom; with the fourth seed on other hardware this is a resolved effect, small. `H_basis64` was +0.010 on one seed at three seeds and is -0.021 at four, its seed 0 never having trained. `H_linear` and `H_state` sit on `H_ref`; the GEARS row is -0.008. The early single-epoch peaks that had `H_state` and `H_pergene` ahead at 14.7 h do not survive the rolling maximum at the matched budget. `H_ref` itself reads 0.174 at 1,400 epochs on the E\_full stack against the v9 band mean of 0.167 at 1,500, the embedding-content effect of the v10 and v11 rounds seen again; the v11 E\_full run at 1,400 on Delta is the same configuration on other hardware and is the direct check when it syncs.



**Figure 7:** Only `H_concat` is above `H_ref` in every seed, by +0.0135 over four seeds, a small resolved effect; three runs never left the plateau. **a**, validation Pearson (5-epoch rolling mean) of all 32 runs, color by arm, line style by seed; the dotted line is the matched budget. **b**, `roll_max` at epochs  $\leq 1,399$  per arm and seed, lines joining one seed's arms; the band is the eight v9 long-budget arms at 1,500 epochs. **c**, the paired difference against `H_ref` within seed, bar at the mean. Generated by `head_round_readout.py`.

### 3.6 Packed five-day runs die of host memory ✗

Four of four packed tasks that ran past three and a half days ended `OUT_OF_MEMORY` (2369697 0–1, 2371531 2–3), at `MaxRSS` 61.4 and 61.1 GB. The Pearson-round packed tasks read 21.3 GB at 1.9 days and the solo batch-64 tasks 17.2 GB. Hypothesis (untested): resident memory grows roughly linearly through a run, in which case the packed Pearson tasks reach the 60 GB line near day five, at the edge of their wall. What grows is not identified; the `file_system` sharing strategy introduced in `f1bfa95b` to stop the file-descriptor exhaustion is the obvious suspect and is unverified. The memory of a running job cannot be raised, and the two rounds that finished lost one run per task to it rather than the whole task.

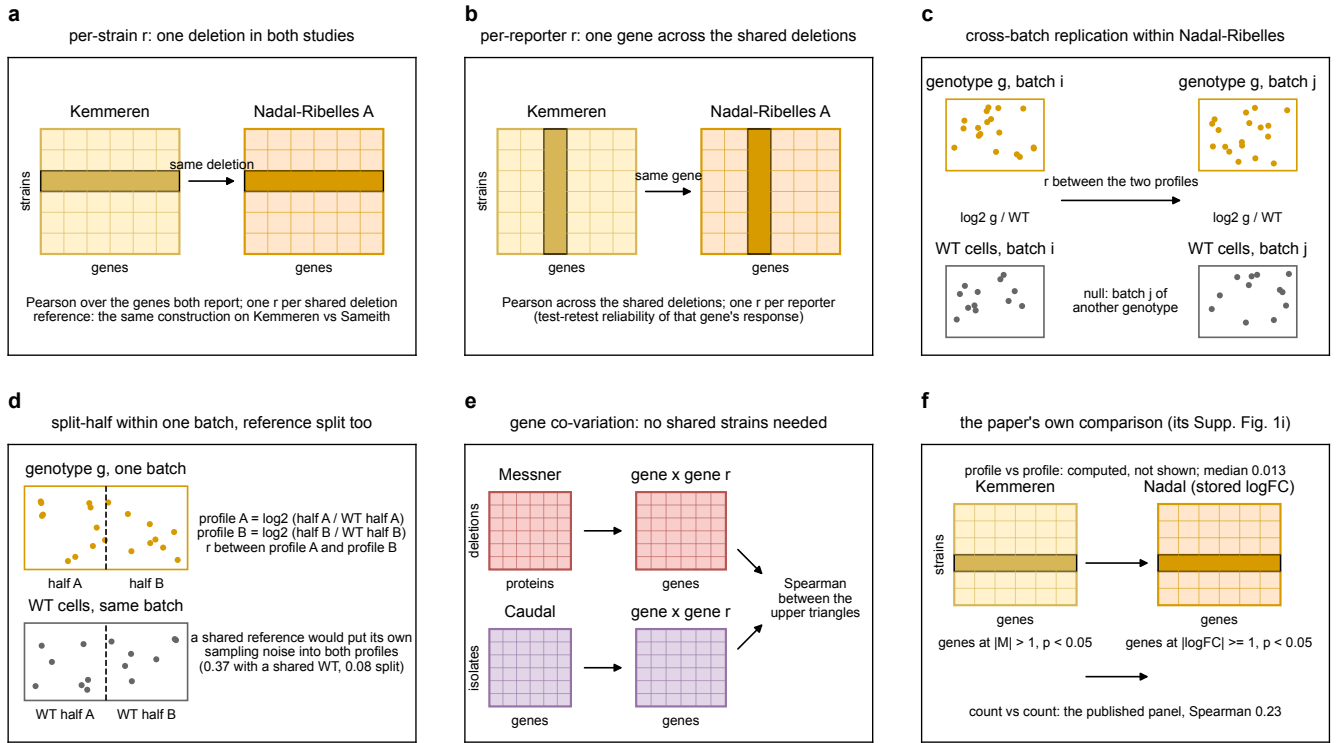
### 3.7 A third deletion panel, and whether it measures the same thing ✗

Every round above trains and scores on the Kemmeren and Sameith microarrays, whose perturbation axis has 1,484 points (section 1.1). Nadal-Ribelles et al. 2025 (doi:10.1038/s41467-025-57600-4) profiled about 3,100 single deletions of the same collection by single-cell RNA-seq, pseudobulked per genotype against a wild type in the same run, in base medium and under 0.4 M NaCl. The base-medium records are 3,039 deletion ORFs, of which 914 are in Kemmeren and 44 in Sameith; the union of the three panels is 3,609 deletions, 2.4 times today’s axis. Before that panel can extend the axis it has to agree with the other two on the deletions they share, and fig. 9 measures that with the same test-retest construction as the ceiling, on the base-medium records only, with Kemmeren against Sameith as the reference for what two studies of one platform look like. Every comparison in this subsection is one of the designs drawn in fig. 8 and listed in table 17: what is held fixed, what one number is computed from, and the reference value the same construction gives on Kemmeren against Sameith. `SRC: experiments/028-knockout-expression/results/cross_study_ko_expression.json`

**Table 17:** The comparisons in this subsection. “Kem”, “Sam”, “Nad”, “Mes”, “Cau” are Kemmeren 2014, Sameith 2015, Nadal-Ribelles 2025 (A, the pseudobulk recompute, unless stated), Messner 2023 and Caudal 2024. The reference column is the same construction on two microarray studies of one platform; a dash means the design has no cross-study reference. Designs are drawn in fig. 8.

| comparison (figure)                      | one number per                  | statistic                                                     | reference, Kem vs Sam                                  | Kem vs Nad            |
|------------------------------------------|---------------------------------|---------------------------------------------------------------|--------------------------------------------------------|-----------------------|
| per-strain profile (9e, 10b)             | shared deletion                 | Pearson over shared genes                                     | 0.74                                                   | 0.003 stored, 0.013 A |
| per-reporter (9f, 10c)                   | shared reporter                 | Pearson across shared deletions                               | 0.62                                                   | 0.030, 0.019          |
| strong responders (9e)                   | deletion                        | Pearson on reporters with $ \text{Kem}  > 1$ ; sign agreement | sign 93 %                                              | 0.10; 60 %            |
| deleted gene’s own rank (11)             | deletion                        | rank of the deleted gene in its profile                       | Kem: rank 1 in 60 %                                    | Nad: rank 1 in < 1 %  |
| cross-batch replication (12a)            | pair of batches of one genotype | Pearson, per-batch WT reference                               | –                                                      | 0.043; null 0.022     |
| split-half replication (12b)             | genotype in one batch           | Pearson, WT split too                                         | –                                                      | 0.080; null 0.051     |
| gene variability (13a)                   | reporter                        | Spearman of sd across shared strains                          | 0.88                                                   | –0.19                 |
| strain similarity (13c)                  | strain pair                     | Spearman of within-study pair $r$                             | 0.52                                                   | 0.03                  |
| GO co-annotation, within one study (13d) | strain pair                     | AUROC, co-annotated vs other pairs                            | Kem responsive 0.55                                    | Nad 0.51 to 0.53      |
| ribosomal protein module (13e)           | deletion                        | Spearman of the module mean                                   | 0.54                                                   | 0.05                  |
| the paper’s Supp. Fig. 1i (12e)          | genotype                        | Spearman of DE-gene counts                                    | –                                                      | 0.23                  |
| the paper’s unshown statistic (12f)      | genotype                        | Spearman of the two profiles                                  | –                                                      | 0.013                 |
| proteome replication (16a)               | deletion                        | Pearson of the two protein profiles, Zelezniak vs Messner     | 0.08 ( $n = 89$ )                                      | –                     |
| gene co-variation (14d–f, 15)            | gene pair                       | Spearman of gene-gene $r$ between panels                      | Kem–Sam 0.68, Kem–Cau 0.47, Mes–Kem 0.30, Mes–Cau 0.34 | 0.03 to 0.10          |

**The stored values are on another scale, and one cell in five is a sentinel.** The microarray  $\log_2$  ratios have standard deviation 0.23 and 0.18; the stored Nadal-Ribelles values have 10.1 as stored and 0.89 after the correction below (fig. 9a, b). They are scanpy `logfoldchanges` from a Wilcoxon rank-sum call, computed as  $\log_2$  of the ratio of two `expm1`-transformed means with a  $10^{-9}$  pseudocount, which returns  $\pm 23$  to  $\pm 33$  whenever one group’s mean is zero: 20.2 % of the stored cells carry such a value and 22.6 % of genes carry it in more than half of their records. The Wilcoxon test is the paper’s DE test; the fold change beside it is a separate quantity, and the paper used it only above



**Figure 8: Only the gene-pair design (e) needs no shared strain, so it is the one comparison that admits the Caudal isolates.** **a**, per-strain: one deletion's profile in each study, correlated over the genes both report. **b**, per-reporter: one gene's response across the shared deletions in each study. **c**, cross-batch replication inside Nadal-Ribelles: one genotype's cells in two cartridge batches, each pseudobulked against the wild-type cells of its own batch; the null pairs batch  $j$  of another genotype. **d**, split-half inside one batch with the wild-type cells split too, so the two profiles share no cell. **e**, gene co-variation: within one panel the correlation of every gene pair across that panel's strains, then the two panels' upper triangles compared, which needs no shared strain and admits the Caudal isolates. **f**, the paper's own comparison: DE-gene counts per genotype (published) and the profile correlation its script computes (not published). Generated by `comparison_designs.py`.

an independent filter at mean normalized counts of 1.27 and a  $p$  threshold, whereas the loader stored it for every gene with a nonzero count. Those cells are treated as absent here. Even after that, the per-strain spread of a Nadal-Ribelles profile is bimodal, with a mode at sd 0.86 against the microarrays' 0.19 and 0.15 (fig. 9c), so the remaining values are still not on the microarray scale.

**On the shared deletions the two panels almost do not agree.** Over 914 shared deletions the per-strain profile correlation with Kemmeren has median 0.003, against 0.744 for Kemmeren with Sameith over 82; the per-reporter test-retest median is 0.030 against 0.620; the OLS slope of Kemmeren on Nadal-Ribelles over 4.06 million shared cells is 0.006 against 0.82 on Sameith (fig. 9d-f). On the reporters where Kemmeren's effect exceeds one  $\log_2$  unit the agreement is weak but above chance, median 0.10 with signs agreeing 60 % of the time (chance 50 %, the microarray pair 93 %). The gene mapping is right: the *cys3* deletion carries MET14, MET3 and MET6 up by 3 to 3.7 and CYS3 itself down, the induction that deletion is known for. The paper's only comparison to Kemmeren is a count of genes at  $p < 0.05$  over 837 shared genotypes; it reports no profile correlation.

**What the agreement tracks.** It rises with the strain's effect size in Kemmeren (Spearman 0.37 over 914 strains) and with the reporter's variance across strains (0.29 over 5,405 reporters), and it does not track the number of cells behind the pseudobulk at all (Spearman 0.02; median 92 cells per genotype) (fig. 9g-i). A pseudobulk that was merely undersampled would agree better where more cells were pooled; this one does not.

**Recomputing the fold change from the cells does not recover agreement.** The single-cell object (`seus_split.RData`, Zenodo 14062629, now in the raw mirror) holds 326,438 control cells over 3,506 genotype labels with raw UMI counts, so the statistic can be replaced. Three were computed per genotype against the 500 pooled wild-type cells: a bulk-like pseudobulk ratio, summed UMI to counts per million,  $\log_2((\text{cpm}_g + 1)/(\text{cpm}_{wt} + 1))$ ; Seurat's `avg_log2FC`; and scanpy's own formula, which reproduces the stored sentinel rate (22.6 % against 20.2 %, cell-by-cell sentinel status agreeing 98 %) and confirms what was stored. Against Kemmeren over 1,002 shared deletions the per-strain median is 0.013, 0.008 and 0.011 for the three, the per-reporter median 0.019, 0.020 and 0.012, and the slope 0.002, 0.010 and 0.000 (fig. 10a-e); none tracks cell count. The statistic was not the cause. SRC: `experiments/028-knockout-expression/results/cross_study_recomputed.json`

**The conditions differ in the medium, and the pooling is brief.** Kemmeren and Sameith grew each strain alone in SC with 2 % glucose at 30 °C in 24-well plates to early mid-log at OD<sub>600</sub> 0.6, two independent cultures each hybridized in dye-swap against a common wild-type reference RNA, with same-day wild-type cultures and a pool of 200 wild-type profiles as the statistical reference, BY4742 MAT $\alpha$  (Sameith follows the same procedure by citation and adds no numbers of its own). Nadal-Ribelles recovered the URA3-marked collection on URA<sup>-</sup> at 25 °C for 48 h to saturation, diluted each strain 1:40 into YPD in 96-well plates for 6 h to OD<sub>660</sub> 0.6 to 0.8 (temperature not stated), pooled the plates into one flask, applied the 15-minute control or NaCl treatment to the pool, and fixed it in methanol; the pooled window is the treatment, not the growth. Cells were captured on Singleron GEXSCOPE microwell cartridges, 14 for the control condition, and the reference is 500 wild-type cells from seven barcoded wild-type strains in the same run. Kemmeren's own SC-against-YPD comparison of wild type found 128 genes changed at 1.7-fold, so the baseline differs, and a deletion's response can differ with it. Hypothesis, unmeasured here: the medium lowers cross-study agreement below the 0.62 same-platform reference. Whether it takes it to zero is what the within-study replication and the cell-level measurements below address, and those do not depend on Kemmeren.

**Read-through from the barcode cassette is not what puts counts on the deleted gene.** The redesigned cassette shortens the heterologous terminator from 262 to 43 nt so that the pTEF1-driven URA3-barcode transcript ends at the deleted gene's own terminator, and 3'-end capture reads that transcript at the deleted locus. If that were the source of the deleted gene's counts in its own cells, those counts would sit at one pTEF1 level whatever wild type expresses there. Measured over 2,724 genotypes with at least 20 cells, the fraction of a genotype's cells detecting its deleted gene tracks the wild-type fraction all the way down (Spearman 0.64) at about 0.6 $\times$  in every bin, and for the 1,586 deleted genes wild type itself detects in under 5 % of cells the genotype's own cells detect it in a median 2.9 %. The counts at the deleted locus scale with native expression; the cassette does not supply them. SRC: `experiments/028-knockout-expression/results/nadal_readthrough_and_signature_tests.json`

**Where a response is detectable, about half the labeled cells carry it; for most genotypes none is detectable.** Two single-cell tests that never use the deleted gene. For the 240 shared deletions with at least 20 Kemmeren reporters beyond one  $\log_2$  unit, every cell of the genotype and every wild-type cell was scored on Kemmeren's signature (mean over up reporters minus mean over down reporters). The score separates the genotype's cells from wild type at a median AUROC of 0.53 (IQR 0.50 to 0.61), 8 of 240 above 0.8; a median 7.5 % of a genotype's cells sit above the wild-type 95th percentile, against 5 % by chance. The 8 that separate are the strong regulators, *rfx1*, *rim101*, *skn7*, *sam1*, *ada2*, *snf5*, *swi3*, YGR122W, and in those 33 to 68 % of the labeled cells sit above the wild-type 95th percentile, the rest inside the wild-type range. The second test asks for any reproducible signal: each genotype's cells were split in half, a 50-gene signature derived from one half against half the wild-type cells, and the other half scored against the held-out wild-type half. Over 207 genotypes with at least 40 cells the median AUROC is 0.52 (IQR 0.49 to 0.57) against a null of 0.49

(wild-type cells playing a genotype), with 18 above 0.65 and 6 above 0.8; the two tests agree on which genotypes separate (Spearman 0.24). SRC: `experiments/028-knockout-expression/scripts/nadal_signature_mixture.R,nadal_split_half_signal.R`

The excess-zero purity estimate (median 0.36 to 0.40 from the deleted gene's own counts; fig. 10f) and the signature fractions above say the same thing for the genotypes with a strong response: roughly half of the cells under a label are that genotype. For the majority of genotypes the labeled cells carry no reproducible signal at this depth (a median of 92 cells at 215 to 7,713 UMI each), and whether that is the same impurity, a response too small for the depth, or a response absent in YPD is not separable here. Batch composition is not the cause; 3,307 of the 3,506 genotypes span three or more of the 14 batches and the wild-type cells span all of them. What produces the impurity in the strong-response genotypes, the clone-and-genotype barcode consensus (a cell called by only one of the two libraries keeps that call), ambient barcode reads in the microwell format, or strain identity in the library, is not measured here.

**The deleted gene cannot be read off its own profile.** A deletion removes its own transcript, so in a clean knockout profile the deleted gene sits at the bottom. In Kemmeren it is the single most reduced gene in 59.5 % of 1,479 strains and in the bottom 1 % in 93.4 %, with a median own value of  $-2.48$ . In Nadal-Ribelles it is at rank 1 in 0.1 to 0.8 % of genotypes and in the bottom 1 % in 2.0 to 8.2 % under the three statistics, with a median rank at the 35th to 47th percentile of the profile (chance is the 50th) and a median own value of  $-0.02$  to  $-0.41$  (fig. 11). The rate does not rise with cells per genotype (2.0 % under 30 cells, 0 % above 300). The deleted gene is not identifiable from a Nadal-Ribelles profile, and read-through from the cassette is not the reason (above). SRC: `experiments/028-knockout-expression/results/nadal_identify_deletion.json`

**The panel does not replicate itself.** The microarray pair has a test-retest of 0.74 per strain; the single-cell panel had none, and without one a cross-study zero cannot be read. The same genotype's cells fall in several of the 14 control cartridges, so its pseudobulk in one batch against that batch's wild-type cells can be compared with its pseudobulk in another (fig. 12a). Over 17,554 pairs of batches of one genotype with at least 15 cells in each, the median Pearson is 0.043; two different genotypes across the same two batches, cell counts matched, give 0.022; 1.8 % of same-genotype pairs exceed 0.2 against 0.2 % of the null. The value does not rise with cells (0.042 at 15 to 30 cells, 0.051 above 120). With the pooled wild type as reference both become 0.11: the batch effect (wild type against wild type between batches, sd 0.75  $\log_2$ , 16 % of genes beyond 2-fold) enters every profile as a common component, and a per-batch reference removes it. Splitting one genotype's cells within one batch in half, with the wild-type cells split too so no reference is shared, gives 0.080 against an other-genotype null of 0.051 over 786 genotype-batches with at least 30 cells (fig. 12b); with a shared reference the same split reads 0.37, which is the reference's sampling noise. A Nadal-Ribelles genotype profile reproduces itself by 0.02 to 0.03 above a different genotype, and 0.04 within the study is what a zero against Kemmeren predicts. SRC: `experiments/028-knockout-expression/scripts/nadal_batch_replication.R`

**Coverage and depth.** A Kemmeren profile carries 6,182 reporters with a ratio and a  $p$  value; Kemmeren calls 52 % of its 1,484 mutants responsive (at least four genes at 1.7-fold and  $p < 0.05$  after excluding 58 wild-type-variable genes; the paper's figure is 47 %), with a median of 4 changed genes. A Nadal-Ribelles cell has a median 934 UMI over 492 genes, and a genotype's own cells put five or more UMI on a median 2,826 of the object's 6,951 genes (120 under 10 cells, 2,558 at 30 to 100, 4,810 above 300); the pseudobulk reports 4,601 because the 500 wild-type cells satisfy the summed-UMI rule on their own (fig. 12c, d). The paper's DEG count per genotype (median 10) is a cell-count artifact: it falls with cells at Spearman  $-0.67$ , because a genotype with few cells has more genes at a zero mean and each is a  $\pm 23$  sentinel that passes  $|\log FC| \geq 1$ . SRC: `experiments/028-knockout-expression/results/nadal_replication_coverage.json`

**What the paper itself computed against the deleteome.** Its released script computes, per genotype with more than five cells, the Spearman between the single-cell fold change and Kemmeren's  $M$  over the shared genes, and the two studies' changed-gene counts; it plots only the counts (its Supplementary Fig. 1i) and the text calls that "a consistent correlation" without a coefficient. Rerun from the released objects over 874 genotypes: the count-against-count Spearman is 0.23 (0.26 with cells partialled out), and the profile correlation it computed and did not show has median 0.013, IQR  $-0.006$  to 0.037, one genotype above 0.2 (fig. 12e, f). The paper's own number is the number measured here. SRC: `experiments/028-knockout-expression/scripts/nadal_paper_deleteome_comparison.R`

**Coarser structure: what the two panels share is small and on the growth axis.** Six tests on the 1,002 shared deletions, each with Kemmeren against Sameith as the same-platform reference (fig. 13). Which genes vary: the reporter's sd across the shared strains agrees between Kemmeren and Sameith at Spearman 0.88 and between Kemmeren and Nadal-Ribelles at  $-0.19$ , and the Nadal-Ribelles sd tracks depth ( $-0.50$  with wild-type counts per million); it is sampling noise. Which strains respond: Kemmeren's changed-gene count against the genotype's cross-batch replication in Nadal-Ribelles, Spearman 0.20 over 260 ( $p = 0.001$ ), small and real. Strain similarity: the strain-by-strain correlation matrices agree between Kemmeren and Sameith at 0.52 (the top 1 % Kemmeren pairs rank at AUROC 0.73 in Sameith) and between Kemmeren and Nadal-Ribelles at 0.03 (AUROC 0.50). Function within one study, with no second study involved: deletions annotated to one GO term of 5 to 100 genes are more similar than other pairs at AUROC 0.55 (cellular component) and 0.53 (process) over Kemmeren's responsive strains, 0.50 and 0.48 over all its strains, and 0.51 to 0.53 in Nadal-Ribelles ( $p = 4 \times 10^{-10}$  for the stored values); a function signal of the size Kemmeren shows with its nonresponders included. The ribosomal protein module (159 genes, the slow-growth

axis): mean  $\log_2$  per strain agrees between Kemmeren and Sameith at 0.54, between Kemmeren and Nadal-Ribelles at 0.05; against the authors' own per-cell iESR score averaged per genotype it is  $-0.11$  ( $p = 6 \times 10^{-4}$ , the expected sign) and against their ribosomal read fraction  $+0.09$ . The best case, deletions Kemmeren calls responsive whose Nadal-Ribelles profile replicates across batches at  $r \geq 0.1$ : 14 strains, median cross-study  $r$  0.078, one above 0.2 (*nmd2*, 0.22). SRC: experiments/028-knockout-expression/results/cross\_study\_structure.json

As released, the panel cannot be pooled with the microarrays as a knockout-expression target, and the reason is now measured rather than inferred: a genotype's profile does not reproduce itself within the study (0.04 across batches against 0.02 for a different genotype), so it cannot reproduce Kemmeren's. What survives is small and sits on the growth axis and on function at the level Kemmeren shows with its nonresponders included. The paper's own profile correlation against the deleteome, which it computed and did not show, is 0.013. Where the labeled cells are half another genotype, a per-cell filter on the deleted gene's own counts (the cells at zero) or a re-assignment from the raw barcode reads would be needed, and the released object carries only the consensus label. Its per-genotype dispersion, a single-cell heterogeneity readout the loader already carries, is a different target and is not affected in the same way.

### 3.8 The knockout proteome against three expression panels ✗

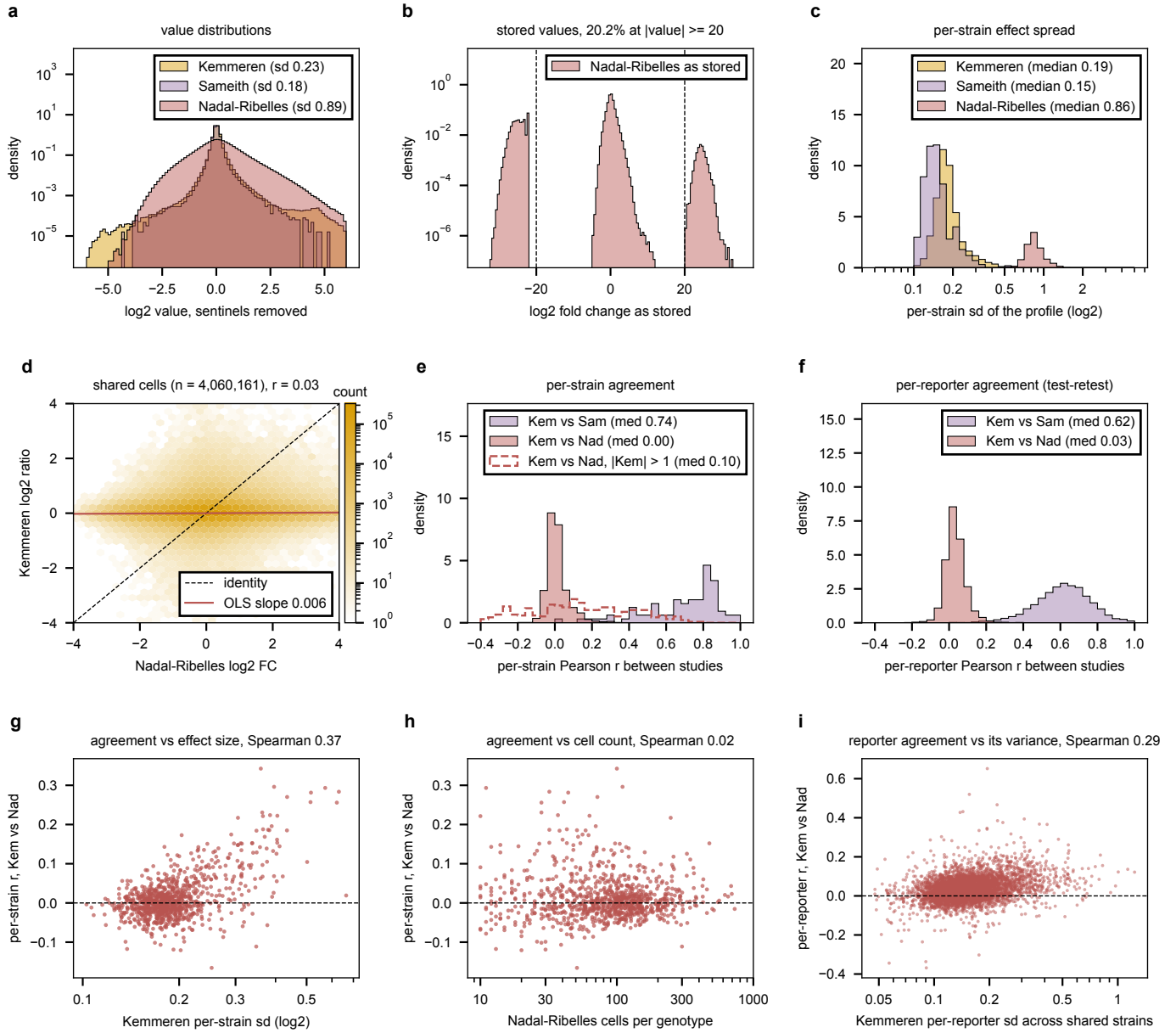
Messner et al. 2023 (doi:10.1016/j.cell.2023.03.026) measured 1,830 proteins over 4,549 deletion ORFs of the same collection by SWATH mass spectrometry in synthetic minimal medium, one measurement per strain against a 388-replicate HIS3 reference. An exploratory pass in July (`proteome_expression_eda.py`, served graph) set it beside Kemmeren strain by strain: per gene across the 1,350 shared deletions a median Pearson of 0.08 (IQR 0.04 to 0.14, 1.7 % of genes above 0.3), per strain across the 1,832 shared genes 0.04 (IQR  $-0.02$  to 0.12), and a ridge map from either modality to the other recovering 2 to 3.5 % of held-out variance, with the medium (SM against SC) named as a confound that attenuates every number. The question here is whether that weak strain-level agreement is what a knockout proteome and a knockout transcriptome share, or whether the two agree at another level, and how the single-cell panel and the natural-isolate panel of Caudal et al. 2024 (943 isolates, RNA-seq) sit against both.

**Strain by strain.** Protein against mRNA  $\log_2$  ratio for the same deletion, z-scored per gene across strains, over the proteins measured in both: with Kemmeren median 0.036 per deletion over 1,350 and 0.075 per protein across deletions, the July numbers reproduced from the LMDs; with Nadal-Ribelles A 0.011 and 0.012 over 2,038 shared deletions (fig. 14a, b). Caudal shares no strain with a deletion panel, so it has no row here.

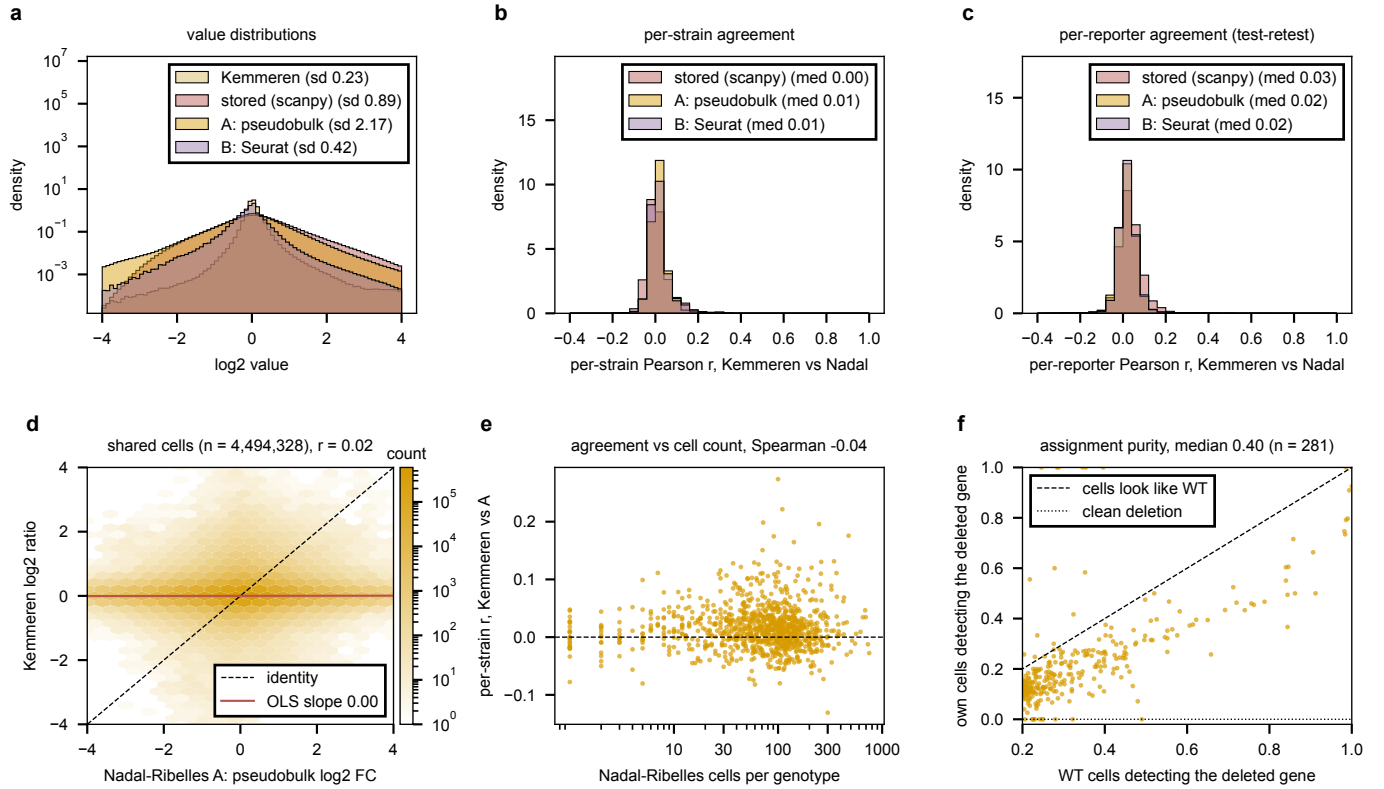
**Gene by gene.** No strain is aligned in this comparison. Within one panel two genes get one Pearson across that panel's own strains, whatever perturbed them, so a panel becomes one gene-by-gene matrix; two panels are compared by the Spearman between their matrices' upper triangles (fig. 8e). It is the only comparison a deletion panel and an isolate panel admit. On the Messner proteins the proteome against the Caudal isolates reads 0.36 over 1,777 genes, with the top 1 % proteome pairs having a median mRNA  $r$  of 0.48 against 0.07 for the rest; against Kemmeren 0.31; Kemmeren against Caudal 0.48. Against Nadal-Ribelles A the three read 0.05, 0.11 and 0.08 (fig. 14d-f). The abundance level itself agrees as expected, protein in the HIS3 reference against mean expression over the isolates at Spearman 0.68 over 1,823 genes (fig. 14c). SRC: experiments/028-knockout-expression/results/proteome\_expression\_covariation.json

**Every panel against every other.** The same comparison over all five panels on one gene list, with the matrices themselves drawn in one ordering (fig. 15). On the 1,645 proteins measured in every panel, Kemmeren against Sameith reads 0.68, Kemmeren against Caudal 0.47, Caudal against Sameith 0.37, and the proteome against the three mRNA panels 0.30, 0.34 and 0.24; Nadal-Ribelles A against any of the four 0.03 to 0.10. On the 4,337 genes measured in every mRNA panel the numbers hold: Kemmeren against Sameith 0.64, against Caudal 0.39, Caudal against Sameith 0.30, Nadal-Ribelles A  $-0.02$  to 0.09. The spread of the gene-pair  $r$  says how much co-variation a panel holds at all: the fraction of pairs beyond  $|r| = 0.3$  is 28 % in Caudal and in Sameith, 16 % in Kemmeren, 7 % in the proteome and 4 % in Nadal-Ribelles A, whose matrix is nearly flat (sd 0.09 against 0.16 to 0.28). Drawn in Kemmeren's cluster order, the two large blocks of Kemmeren appear at the same place in Caudal, in Sameith and, fainter, in the proteome, and nowhere in Nadal-Ribelles A. SRC: experiments/028-knockout-expression/results/gene\_covariation\_all.json

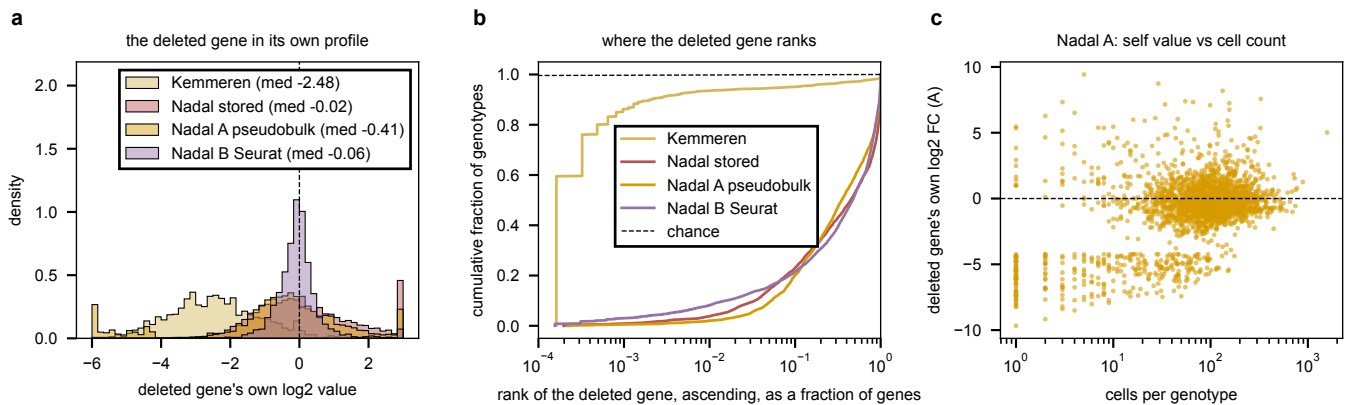
**A second knockout proteome.** Zelezniak et al. 2018 (doi:10.1016/j.cels.2018.08.001), from the same laboratory five years earlier, measured 726 proteins in 97 kinase deletions of the collection, three replicates against a 12-replicate wild type, in SM medium. Messner covers 89 of them, Kemmeren 94, all three 87, so the proteome gets the replication check the transcriptome had with Sameith. On log ratios over each study's own wild type, z-scored per protein: Zelezniak against Messner reads a median 0.08 per deletion and 0.09 per protein (7 % of proteins above 0.3); Zelezniak against Kemmeren mRNA 0.01 and 0.03; Messner against Kemmeren on the same 87 strains 0.01 and 0.02 (fig. 16a, b). The two wild types agree on abundance at Spearman 0.77 over 714 proteins (c). Gene co-variation across the 87 strains is 0.15 between the proteomes and 0.07 against Kemmeren, where Messner's own 87-strain matrix reproduces its full 4,549-strain matrix at 0.80 (d-f): the strain count is enough to recover a panel's structure, and the 0.15 is disagreement between the two studies. SRC: experiments/028-knockout-expression/results/proteome\_replication\_zelezniak.json



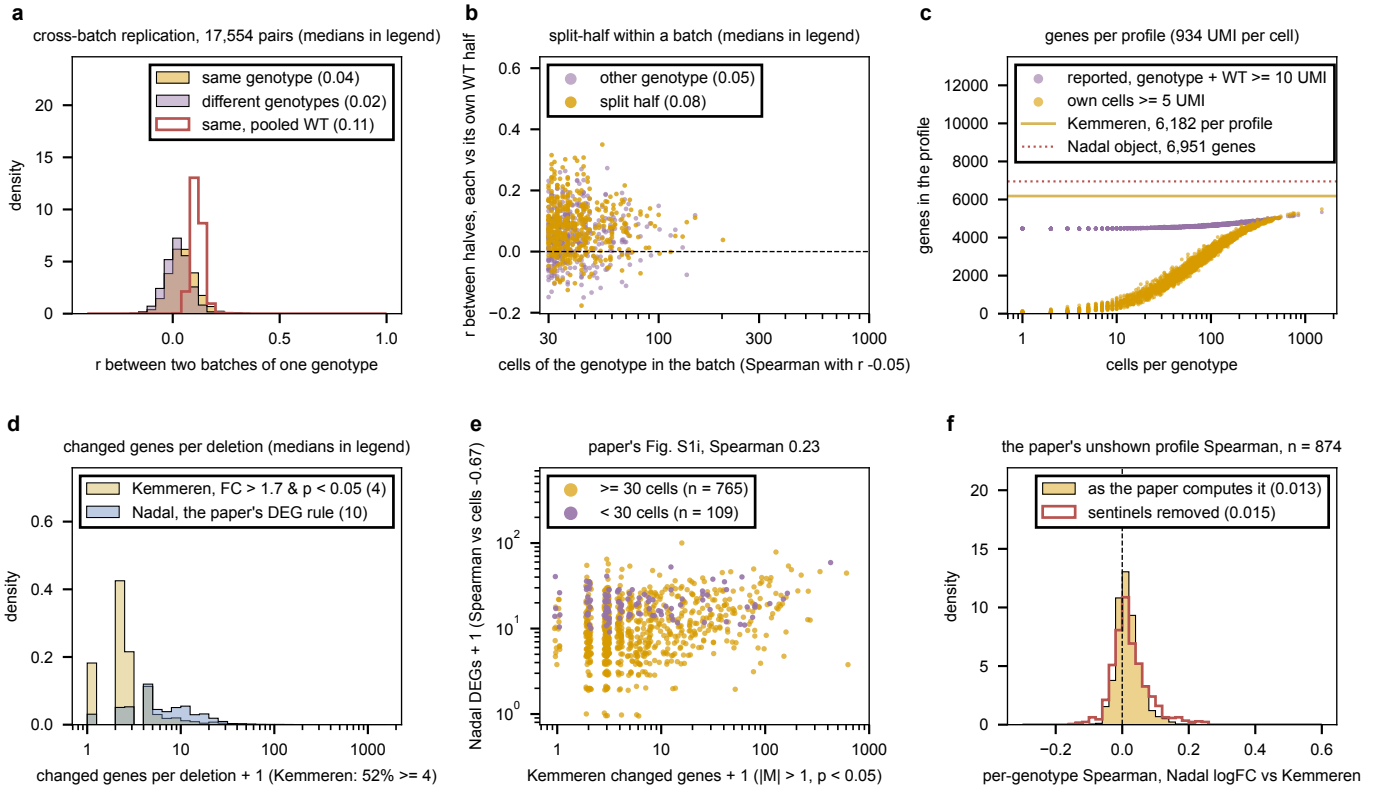
**Figure 9: A deletion's Kemmeren profile reproduces in Sameith (median  $r$  0.74, e) and not in Nadal-Ribelles (0.00), whose values sit on another scale with 20% sentinels (b).** a, log<sub>2</sub> values of each panel with the Nadal-Ribelles sentinels removed; the two platforms sit on different scales. b, Nadal-Ribelles values as stored, with the  $\pm 20$  lines: 20.2% of cells are the zero-mean sentinel. c, per deletion, the standard deviation of its profile across reporters. d, Kemmeren against Nadal-Ribelles on every shared (strain, gene) cell, log-count density, with the identity and the fitted slope. e, per shared deletion, the Pearson across reporters between its two profiles: Kemmeren vs Sameith ( $n = 82$ ), Kemmeren vs Nadal-Ribelles ( $n = 914$ ), and the latter over only the reporters with  $|\log_2| > 1$  in Kemmeren (dashed,  $n = 368$  strains). f, per reporter, the Pearson across shared strains between its two response vectors, the test-retest reliability behind the ceiling. g, the per-strain agreement of e against the strain's Kemmeren spread; h, against the number of cells behind its pseudobulk; i, the per-reporter agreement of f against the reporter's Kemmeren spread across the shared strains. Generated by `cross_study_ko_expression.py`.



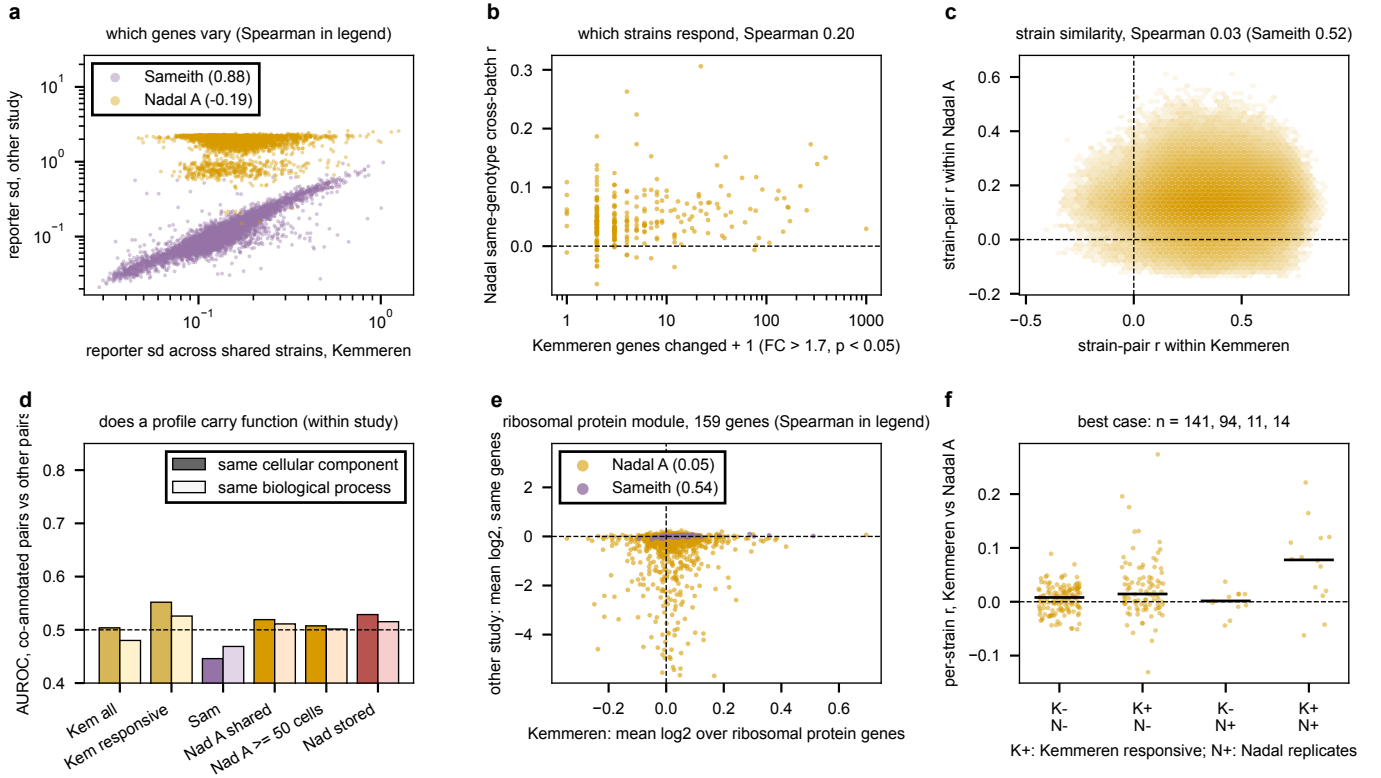
**Figure 10: Recomputing the fold change from the raw cells does not recover agreement with Kemmeren (b, c), because the labeled cells are only about 40 % the genotype they carry (f). a**, log<sub>2</sub> values of Kemmeren and of three Nadal-Ribelles versions: as stored, the bulk-like pseudobulk ratio (A) and Seurat's `avg_log2FC` (B). **b**, per shared deletion, the Pearson across reporters with Kemmeren for each version; **c**, the same per reporter across shared strains. **d**, Kemmeren against A on every shared cell. **e**, per-strain agreement of A against the cells behind the genotype. **f**, assignment purity: for each genotype whose deleted gene wild type detects in at least a fifth of cells, the fraction of the genotype's own cells detecting that gene against the wild-type fraction; the dashed identity is what cells that are not the genotype look like, the dotted zero a clean deletion. Generated by `cross_study_recomputed.py` from `nadal_pseudobulk_recompute.R` and `nadal_assignment_purity.R`.



**Figure 11: The deleted gene is the most reduced gene of its own profile in 60 % of Kemmeren deletions and under 1 % of Nadal-Ribelles genotypes (b). a**, the deleted gene's own log<sub>2</sub> value per genotype for Kemmeren and three Nadal-Ribelles statistics. **b**, cumulative distribution of the deleted gene's ascending rank as a fraction of genes in the profile, log axis; the dashed diagonal is chance. **c**, for the pseudobulk statistic, the deleted gene's own value against the cells behind the genotype; the band near -5 to -8 at low counts is the pseudocount floor of a near-empty pseudobulk. Generated by `nadal_identify_deletion.py`.



**Figure 12: A Nadal-Ribelles genotype barely reproduces itself across batches (0.04 against a 0.02 null, a), its coverage is set by cell count (c), and the paper's own unshown profile correlation with the deleteome is 0.01 (f).** **a**, Pearson between two batches of one genotype (pseudobulk against the batch's own wild-type cells), the different-genotype null with matched cell counts, and the same-genotype value with a pooled wild-type reference. **b**, split-half within one batch, each half against its own half of the wild-type cells, against cells per genotype; gray, the other-genotype null. **c**, genes per profile: covered by the genotype's own cells at five or more UMI, and reported under the summed-UMI rule; lines at Kemmeren's reporters per profile and at the genes in the single-cell object. **d**, changed genes per deletion under each study's own rule, Nadal-Ribelles in blue here so it separates from Kemmeren's yellow. **e**, the paper's Supplementary Fig. 1i reproduced from its released objects, shaded by cells per genotype. **f**, the per-genotype profile Spearman the paper's script computes and does not show, with and without the sentinel genes. Generated by `nadal_replication_coverage.py` from `nadal_batch_replication.R` and `nadal_paper_deleteome_comparison.R`.

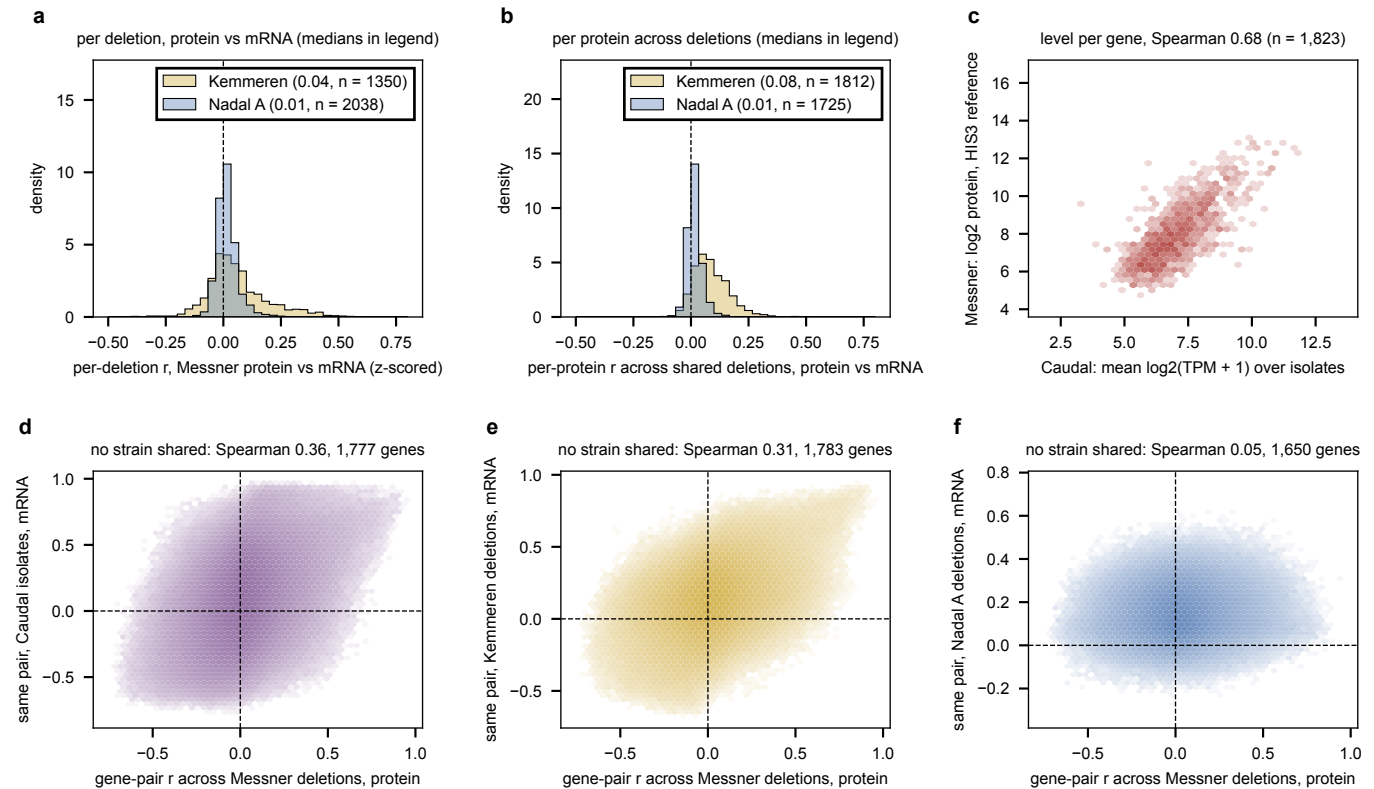


**Figure 13: What carries over from Kemmeren to Nadal-Ribelles is small and on the growth axis, which strains respond (b) and a weak function signal (d); which genes vary (a), strain similarity (c) and the ribosomal module (e) do not.** On the 1,002 shared deletions. **a**, a reporter's sd across the shared strains, Kemmeren against Sameith and against Nadal-Ribelles A. **b**, Kemmeren's changed-gene count per deletion against the genotype's cross-batch replication in Nadal-Ribelles. **c**, strain-pair correlation within Kemmeren against the same pair within Nadal-Ribelles A. **d**, within one study, the AUROC of the pair correlation for pairs annotated to one GO cellular-component (dark) or biological-process (light) term of 5 to 100 genes against other pairs. **e**, mean log<sub>2</sub> over 159 cytosolic ribosomal protein genes per strain, Kemmeren against Nadal-Ribelles A and against Sameith. **f**, per-strain cross-study  $r$  by whether Kemmeren calls the deletion responsive (K+) and whether its Nadal-Ribelles profile replicates across batches at  $r \geq 0.1$  (N+); bars are medians. Generated by `cross_study_structure.py`.

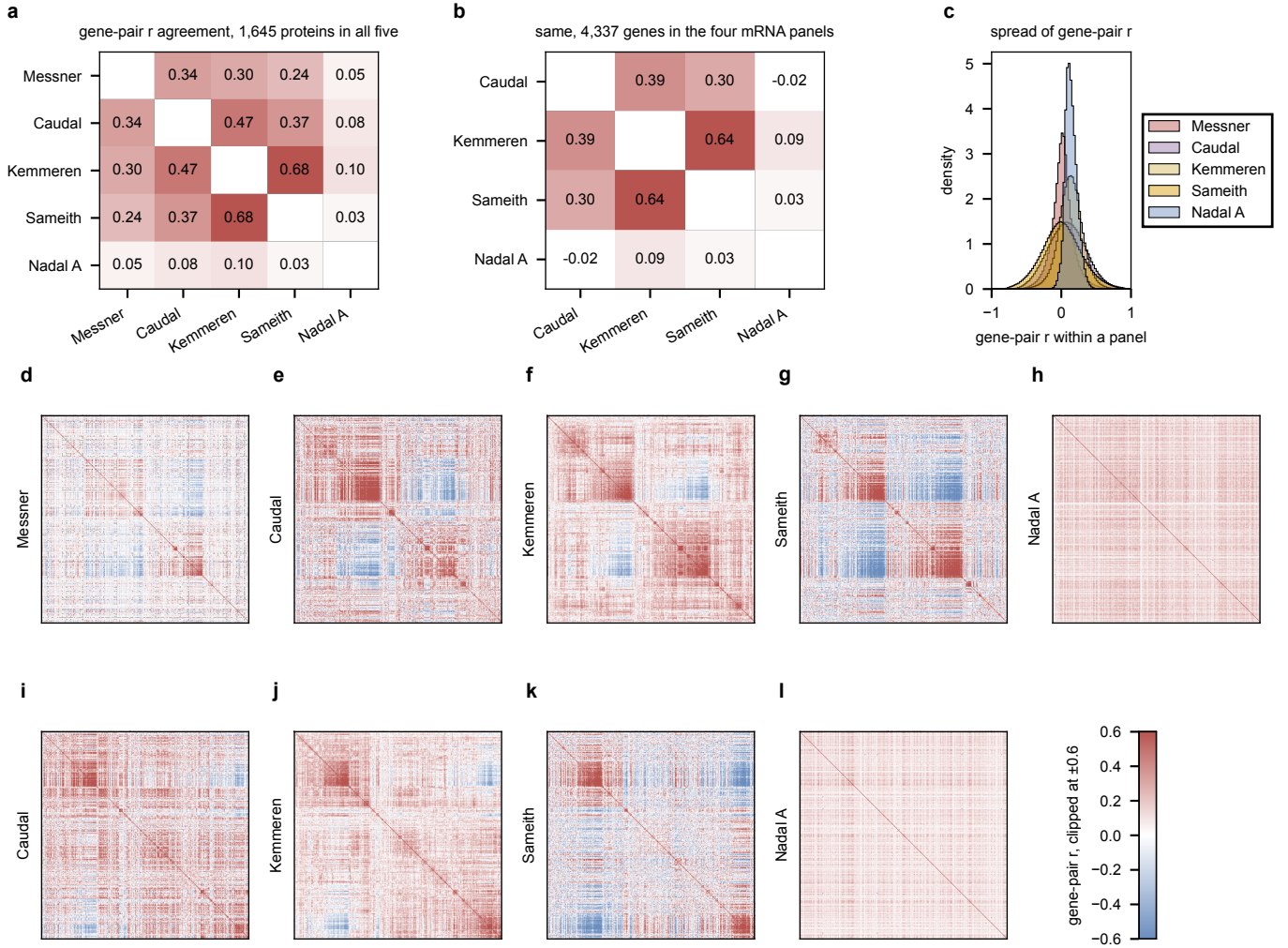
**What differs between the two proteomes.** Both draw on the Euroscarf MATa collection with prototrophy restored by the same minichromosome, in SM medium without supplements at 30 °C, and neither paper says its stocks were checked against the other's. From there the methods part: Zelezniak grew 30 ml shake cultures and sampled each strain “at an OD600  $1.5 \pm 0.1$ , before the cultures enter the diauxic shift”, in three biological replicates, on a TripleTOF 5600 with Spectronaut and a peptide-level surrogate-variable batch correction, against the parental strain; Messner grew 1.6 ml deep-well cultures with a glass bead at 1,000 rpm and harvested every strain 8 h after a 1/10 dilution with no OD control, one measurement per strain (“Strains were not measured in replicates”), on a TripleTOF 6600 with DIA-NN and a plate-median batch correction across 57 plates, two instruments and 12 months, against a HIS3-complemented *his3Δ*. Messner reports a growth rate of  $< 0.8$  for 748 of 4,678 strains, that a random forest on the proteome predicts growth at  $R^2 = 0.68$ , and a per-protein CV of 11.3 % in wild type and 16.2 % across knockouts; Zelezniak reports that “less than 10 % of the total proteome changes in our kinase knockout strains could be explained by changes in growth rate”. Messner’s own re-measurement of 29 deletions in the auxotrophic SGA background, same laboratory and medium, gave per-deletion Spearman “from  $\rho = 0.72$  for the *dep1Δ* deletion strains to  $\rho = -0.19$  for the *paf1Δ* deletion strain”, and Zelezniak’s comparison of its kinase proteomes to microarrays of the same deletions gave a median per-strain  $r$  of 0.12. Hypotheses, untested and in this order: a single unreplicated sample per strain at a 16 % CV bounds a per-strain profile correlation far below the 388-sample abundance agreement; the clock-timed harvest puts slow-growing kinase deletions at a different growth phase than the OD-timed one, on the axis Messner says explains most of its proteome and Zelezniak says explains little of theirs; the surrogate-variable correction in Zelezniak may remove genotype-correlated structure; stock drift (Messner finds 92 aneuploid strains, one of them the kinase *dbf2*, and 44 strains still expressing the deleted gene) is real but touches too few of the 89 to move a median. Media differ least. Neither paper compares itself to the other.

SRC: torchcell-library/messnerProteomicLandscapeGenomewide2023/paper.md, zelezniakMachineLearningPredicts2018/paper.md

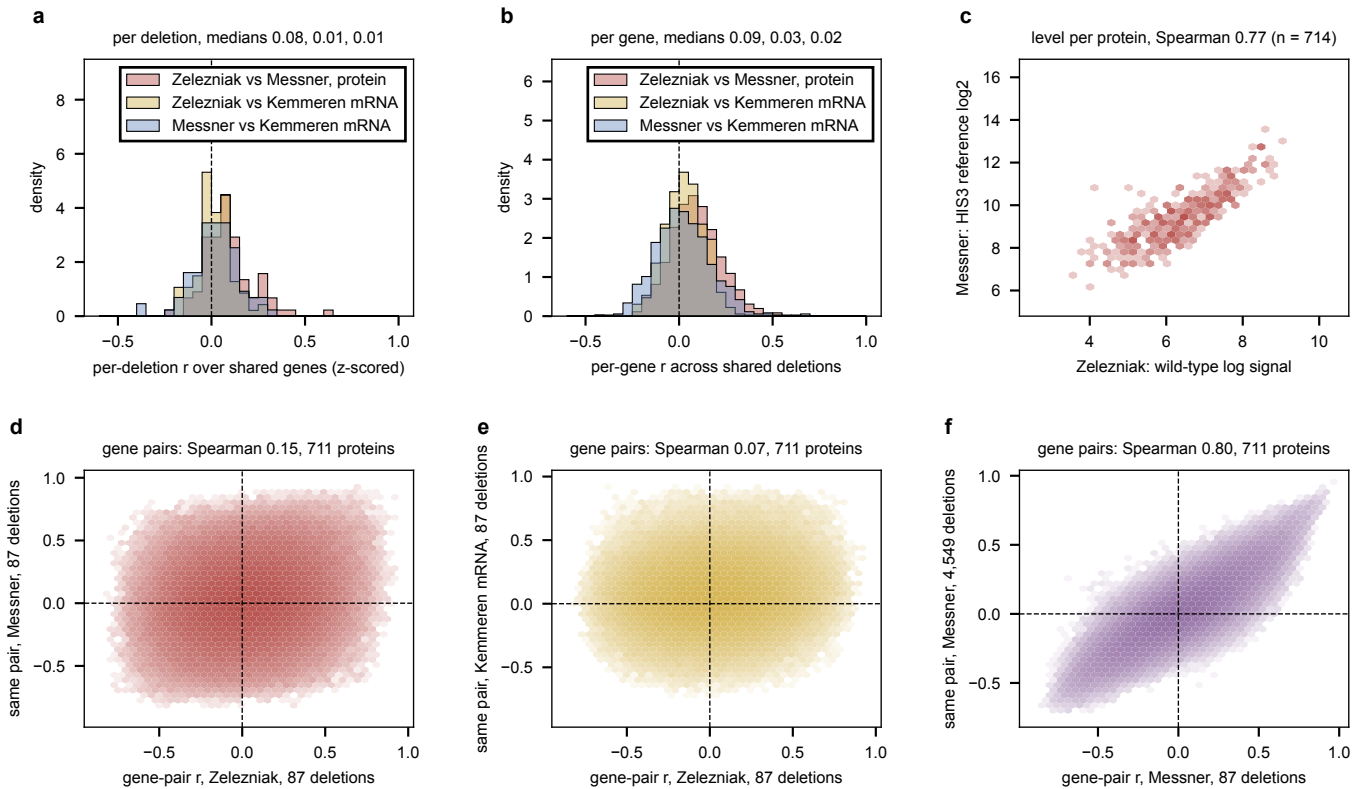
Four panels that differ in platform, perturbation type and medium agree pairwise on which genes co-vary, at 0.24 to 0.68, while the strain-level agreement between the two that share strains is 0.04, and a second proteome on 89 of the same deletions reproduces Messner at 0.08; what a deletion does to a transcript and to its protein is largely different, but the gene modules that move together are the same modules, and the same modules move across natural isolates. The single-cell panel agrees with none of the four at either level and holds little co-variation of its own. For the campaign this is the reason to carry the proteome as a target of its own rather than as a proxy for expression, and to leave the single-cell panel out of both.



**Figure 14: Protein and mRNA barely correlate for the same deletion (a, b, medians 0.04 or less), yet the gene pairs that co-vary in the proteome co-vary in Caudal and in Kemmeren (d, e) and not in Nadal-Ribelles (f).** a, per shared deletion, the Pearson between its protein and mRNA log<sub>2</sub> ratios (z-scored per gene across strains) for Kemmeren and for Nadal-Ribelles A. b, per protein, the Pearson across shared deletions. c, abundance level per gene, protein in the HIS3 reference against mean expression over the Caudal isolates. d–f, gene-pair co-variation: the correlation of two genes across Messner’s deletions (protein) against the same pair across Caudal’s isolates, Kemmeren’s deletions and Nadal-Ribelles A’s deletions (mRNA), log-count density. Generated by `proteome_expression_covariation.py`.



**Figure 15: Four panels that differ in platform, perturbation and medium agree on which gene pairs co-vary (0.24 to 0.68, a, b), and Nadal-Ribelles agrees with none, its matrix nearly flat (h, l). Within a panel, two genes get one Pearson across that panel's own strains; no strain is aligned between panels. a, Spearman between the upper triangles of two panels' gene-by-gene matrices on the 1,645 proteins measured in all five panels. b, the same on the 4,337 genes measured in the four mRNA panels. c, the distribution of the gene-pair  $r$  within each panel on the protein gene list. d-h, the five matrices on the protein gene list, and i-l the four mRNA matrices on the expression gene list, every matrix in the ordering given by average-linkage clustering of Kemmeren's matrix on that gene list, so a block in Kemmeren can be looked for at the same place in every other panel; color clipped at  $\pm 0.6$ . Messner spans 4,549 deletions, Caudal 943 isolates, Kemmeren 1,484 deletions, Sameith 82 single deletions, Nadal-Ribelles A the 2,243 deletions with at least 50 cells. Generated by `gene_covariation_all.py`.**



**Figure 16: Two knockout proteomes from one laboratory agree on how much of each protein is made (0.77, c) and barely on what a kinase deletion does to it (0.08 per deletion, a). Zelezniak 2018 against Messner 2023 on 89 shared kinase deletions, and each against Kemmeren mRNA. a, per shared deletion, the Pearson between the two log ratios over the proteins both measure, z-scored per protein. b, per protein, across the shared deletions. c, wild-type abundance per protein in the two studies. d–f, gene-pair co-variation on 711 proteins across the 87 deletions in all three studies: Zelezniak against Messner, Zelezniak against Kemmeren mRNA, and Messner on the 87 against Messner on all 4,549 deletions, which is what the same panel at that strain count looks like. Generated by `proteome_replication_zelezniak.py`.**

## 4 The linear and neighbor baselines, on the partitions the model trains on $\times$

The four baselines of table 5 were scored on a private permutation of the raw Kemmeren panel, so baseline and trained arm shared a task but not a partition. Everything in this section reads the partition file the training run reads: for split seed  $k$  the `CellDataModule` writes `index_details_seed_k.json`, the record indices per split, and both the baseline script and the GPU launcher take that file as given. The files are hashed in `split_indices_manifest.json`; seeds 0 to 2 carried the same hash on both machines and seed 3 was built once and copied. The trained arms on the same four partitions are the v13 round (section 5), running as this is written; nothing about them is reported here.

**Setup.** Strain  $b$  carries deletion set  $S_b$  and a profile  $y_b \in \mathbb{R}^F$  over  $F = 6,169$  reporters, each entry the  $\log_2$  ratio to the wild-type reference. The partition  $(\mathcal{T}, \mathcal{V}, \mathcal{E})$ , train, validation, test, is the datamodule’s record-level 80/10/10 draw for split seed  $k$ : a seeded shuffle within each index key (phenotype label, perturbation count), the per-key splits intersected. Every deleted gene appears once in the panel, so every strain in  $\mathcal{V}$  and  $\mathcal{E}$  is an unseen perturbation. Per-gene training mean  $\mu_f = |\mathcal{T}|^{-1} \sum_{b \in \mathcal{T}} y_{bf}$  and residual  $r_{bf} = y_{bf} - \mu_f$ . The metric is the training metric,

$$\text{pf}(\hat{Y}, Y) = \frac{1}{|\mathcal{F}_{\text{ok}}|} \sum_{f \in \mathcal{F}_{\text{ok}}} \frac{\langle \hat{y}_{\cdot f} - \bar{\hat{y}}_f, y_{\cdot f} - \bar{y}_f \rangle}{\|\hat{y}_{\cdot f} - \bar{\hat{y}}_f\| \|y_{\cdot f} - \bar{y}_f\|}, \quad \text{nmse}(\hat{Y}, Y) = \frac{\sum_{b,f} (\hat{y}_{bf} - y_{bf})^2}{\sum_{b,f} (y_{bf} - \mu_f)^2}, \quad (2)$$

with the columns and the means taken over the strains of the scored split and  $\mathcal{F}_{\text{ok}}$  the reporters whose prediction and target both have a norm above  $10^{-8}$ . A prediction constant across strains has no surviving column and scores exactly 0, which is what makes the first two baselines a self-check rather than a competitor.

**B0 and B1.**  $\hat{y}_{bf} = \mu_f$  and  $\hat{y}_{bf} = 0$ . Both are 0 on eq. (2) on every partition; B0’s `nmse` is 1 by construction, and the script raises if either lands elsewhere. B1’s `nmse` is how far the panel sits from the reference: 1.33, 1.33, 1.37 and 1.39 on the validation strains of splits 0 to 3.

**B2, the bilinear ridge.** The perturbation is represented by an external gene embedding,  $p_b = |S_b|^{-1} \sum_{g \in S_b} e_g \in \mathbb{R}^D$ , standardized column by column on the training strains; a strain with no embedded gene is excluded from the score and counted. The gene basis  $G \in \mathbb{R}^{F \times K}$  holds the top- $K$  right singular vectors of the training residual matrix  $R_{\mathcal{T}}$ , and

$$W^* = \underset{W}{\text{argmin}} \|R_{\mathcal{T}}G - P_{\mathcal{T}}W\|_F^2 + \lambda \|W\|_F^2 = (P_{\mathcal{T}}^{\top} P_{\mathcal{T}} + \lambda I)^{-1} P_{\mathcal{T}}^{\top} R_{\mathcal{T}} G, \quad \hat{Y} = P W^* G^{\top} + \mathbf{1} \mu^{\top}. \quad (3)$$

This is the benchmark’s  $Y \approx G W P^{\top} + b$  with the intercept removed by centering.  $K \in \{1, 2, 4, 8, 16, 32, 33, 64, 128, 256\}$  and  $\lambda \in \{10^{-3}, \dots, 10^2\}$ , sixty cells, chosen on validation.

**B3, the neighbor mean.** With  $s_{bj} = \cos(p_b, p_j)$  over training strains  $j$  and  $N_k(b)$  the  $k$  largest,  $\hat{r}_b = k^{-1} \sum_{j \in N_k(b)} r_j$  and  $\hat{y}_b = \hat{r}_b + \mu$ ,  $k \in \{1, 3, 5, 10, 25\}$  chosen on validation. No fitted parameter.

**The kNN probe, the same idea on raw profiles.** The earlier probe (`knn_embedding_probe.py`) predicts the whole profile rather than the residual, weights the neighbors by their clipped similarity, and ran on the split-0 partition file before any of this existed, so it is already on the model’s partition:

$$\hat{y}_b = \sum_{j \in N_k(b)} w_{bj} y_j, \quad w_{bj} = \frac{[s_{bj}]_+}{\sum_{j' \in N_k(b)} [s_{bj'}]_+}, \quad s_{bj} = \cos(e_b, e_j), \quad (4)$$

over fourteen embeddings and random embeddings of matched dimension as the floor. Its validation set is the 140 single-deletion strains of split 0.

**Selection mirrors the model’s.** A trained arm reports the epoch that maximizes validation Pearson, an optimistic order statistic over epochs. B2’s cell and B3’s  $k$  are likewise the validation maximum, over sixty and five cells, so the validation columns below carry the same kind of optimism as `roll_max`, and the test columns are the read that does not. The 90/10 column folds the test records into train against the same validation strains, the linear analog of the round’s 90/10 arm, and has no test read.

**The partition moves a linear model as much as it moves the network.** B2 on ProtT5 reads 0.135, 0.076, 0.117 and 0.094 on the four validation draws, a spread of 0.026 from a sixty-cell ridge with no seed of its own, and B3 on ProtT5 spreads 0.051 on test. The four draws share the same training count and differ in which 155 strains are held out (the 72 Sameith doubles land 15, 13, 7 and 10 in validation), so the spread is the partition itself. Eight more partitions, split seeds 4 to 11, exist only for the linear models, and over the twelve draws B2 on ProtT5 reads  $0.104 \pm 0.018$  on validation (range 0.076 to 0.135) and  $0.108 \pm 0.017$  on test; B3 on ProtT5  $0.122 \pm 0.020$  and  $0.118 \pm 0.034$ . Split 0 is the highest of the twelve validation draws for B2 and the third highest for B3. That is the axis the v13 round

**Table 18:** B2 and B3 on the four `CellDataModule` partitions the v13 round trains on, `pearson_per_feature` on val and on test, where rank, ridge and  $k$  are chosen on val and test is read at the chosen cell. Each partition has 1244 training strains and 155, 151, 155, 155 val (155, 150, 155, 155 test) labeled expression strains. The last column is split 0 with the test records folded into train (1399 training strains, the same val set), the linear analog of the round’s 90/10 arm. B0 is 0 and B1 is 0 on every partition by construction; B1’s val nmse is 1.330, 1.327, 1.372, 1.391 on splits 0 to 3.

SRC: `experiments/019-simb-multimodal/scripts/expression_baselines_split.py`

| baseline          | embedding          | read | split seed |        |        |        | mean $\pm$ sd     | 90/10  |
|-------------------|--------------------|------|------------|--------|--------|--------|-------------------|--------|
|                   |                    |      | 0          | 1      | 2      | 3      |                   |        |
| B2 bilinear ridge | ProtT5             | val  | +0.135     | +0.076 | +0.117 | +0.094 | $0.106 \pm 0.026$ | +0.137 |
|                   |                    | test | +0.085     | +0.148 | +0.111 | +0.097 | $0.110 \pm 0.027$ | –      |
| B2 bilinear ridge | CaLM               | val  | +0.098     | +0.081 | +0.080 | +0.099 | $0.089 \pm 0.010$ | +0.102 |
|                   |                    | test | +0.069     | +0.120 | +0.087 | +0.093 | $0.092 \pm 0.021$ | –      |
| B2 bilinear ridge | species LM         | val  | +0.043     | +0.044 | +0.022 | +0.042 | $0.038 \pm 0.010$ | +0.044 |
|                   |                    | test | +0.061     | +0.019 | +0.017 | +0.060 | $0.040 \pm 0.024$ | –      |
| B2 bilinear ridge | chromatin pathways | val  | +0.107     | +0.032 | +0.036 | +0.036 | $0.053 \pm 0.036$ | +0.110 |
|                   |                    | test | +0.004     | +0.040 | +0.031 | +0.006 | $0.020 \pm 0.018$ | –      |
| B3 neighbor mean  | ProtT5             | val  | +0.130     | +0.113 | +0.111 | +0.126 | $0.120 \pm 0.009$ | +0.141 |
|                   |                    | test | +0.046     | +0.162 | +0.124 | +0.142 | $0.118 \pm 0.051$ | –      |
| B3 neighbor mean  | CaLM               | val  | +0.122     | +0.092 | +0.072 | +0.117 | $0.101 \pm 0.023$ | +0.123 |
|                   |                    | test | +0.088     | +0.115 | +0.086 | +0.111 | $0.100 \pm 0.015$ | –      |
| B3 neighbor mean  | species LM         | val  | +0.061     | +0.101 | +0.029 | +0.056 | $0.062 \pm 0.030$ | +0.070 |
|                   |                    | test | +0.045     | +0.050 | +0.041 | +0.056 | $0.048 \pm 0.006$ | –      |
| B3 neighbor mean  | chromatin pathways | val  | +0.063     | +0.020 | +0.016 | +0.032 | $0.033 \pm 0.021$ | +0.058 |
|                   |                    | test | –0.008     | +0.011 | +0.003 | +0.003 | $0.002 \pm 0.008$ | –      |

**Table 19:** The parameter-free neighbor probe on expression: a held-out deletion’s profile predicted as the similarity-weighted mean of the profiles of its  $k$  nearest other deleted genes in the embedding space named, `pearson_per_feature` on the 140 single-deletion validation strains of split seed 0, best  $k$  over  $\{1, 3, 5, 10, 25, 50\}$  and three fixed  $k$ . Dimension in parentheses. The random rows are embeddings of matched dimension drawn once, and set the floor. One-hot is undefined (every cosine similarity is zero) and is omitted. SRC: `experiments/019-simb-multimodal/scripts/knn_embedding_probe.py`

| embedding (dim)                             | best $k$ | at best $k$ | $k=1$  | $k=10$ | $k=25$ |
|---------------------------------------------|----------|-------------|--------|--------|--------|
| ProtT5 (1,024)                              | 25       | +0.117      | +0.065 | +0.091 | +0.117 |
| ESM2 650M (1,280)                           | 50       | +0.092      | +0.064 | +0.080 | +0.083 |
| CaLM (768)                                  | 25       | +0.069      | +0.056 | +0.069 | +0.069 |
| codon frequency (64)                        | 50       | +0.064      | +0.023 | +0.058 | +0.046 |
| chromatin pathways (197)                    | 3        | +0.051      | +0.015 | +0.021 | +0.024 |
| species LM, both flanks (1,536)             | 25       | +0.014      | +0.009 | +0.005 | +0.014 |
| nucleotide transformer, both flanks (5,120) | 50       | +0.012      | –0.017 | +0.004 | +0.009 |
| random (10)                                 | 3        | +0.011      | –0.000 | +0.003 | –0.006 |
| random (100)                                | 5        | +0.018      | +0.014 | +0.016 | –0.011 |
| random (1,000)                              | 5        | +0.033      | +0.008 | +0.016 | –0.001 |

was built to measure for the trained arms, and the linear numbers say it is large before any network is involved.  
SRC: experiments/019-simb-multimodal/results/expression\_baselines\_split/summary.json

**Split 0 is the easiest validation draw for the linear models too.** On split 0 both ProtT5 baselines post their highest validation score of the four draws and B2 its lowest test score, 0.085 against 0.148, 0.111 and 0.097 on the others; B3 on ProtT5 drops from 0.130 on validation to 0.046 on test there. Every score the campaign has quoted is a split-0 validation number. The v12 reference stack on that draw reads  $0.174 \pm 0.008$  and  $H_{\text{concat}} 0.188 \pm 0.005$  at 1,399 epochs (section 3.5), against 0.135 for the best linear cell and 0.130 for the best neighbor rule on the same 155 strains, so the margin over the linear model on this partition is 0.04 to 0.05, not the 0.09 that table 5 gave on a different permutation. The earlier 0.104 was not wrong; it was another draw.

**Ten percent more training data buys the linear models almost nothing.** With the 155 test strains folded into train, B2 on ProtT5 moves from 0.135 to 0.137 and B3 from 0.130 to 0.141 on the same validation set, and CaLM by +0.004 and +0.001. Whether the trained arms move more is what task 2 of the round measures; the linear result is the floor that comparison is read against.

**Embeddings rank the same way under every rule.** ProtT5 leads B2, B3 and the probe on every partition; CaLM is second; the species language model over both flanks sits at 0.04 to 0.06; the chromatin-pathway feature fits validation on split 0 and transfers to test at 0.004 and 0.02, the one case where the sixty-cell selection is visibly optimistic. In the probe, every nucleotide-transformer arm is at or below the random floor of 0.033, and ProtT5's 0.117 at  $k = 25$  is what section 1.5 quotes.

## 4.1 Every gene representation, on the proteome and on expression ✗

The gate above reads four representations. The v14 proteome round raised the question of which representation of the deleted gene carries its protein response, and whether a protein language model other than ProtT5, a DNA model, or a composite would move the linear floor. So every representation the embedding builder serves was run as the perturbation input of B2 and B3 on the same four partitions, for the proteome build (3,581 training strains, 1,536 proteins after the coverage filter) and for the expression build: two protein language models (ProtT5 and ESM2 650M, each with and without the dubious ORFs), two coding-sequence representations (CaLM, codon frequency), the species-aware fungal transformer over the 5' and 3' flanks, the Nucleotide Transformer over six windows, the chromatin-pathway graph vector, random controls at three widths, and eight composites including the four-embedding stack the trained model consumes. table 20 and fig. 17 carry every number; the seed sd is 0.005 to 0.012 on the proteome and 0.01 to 0.03 on expression, so a difference under about 0.02 between two rows is not resolved by four partitions.

SRC: experiments/019-simb-multimodal/results/baselines\_embedding\_study.json

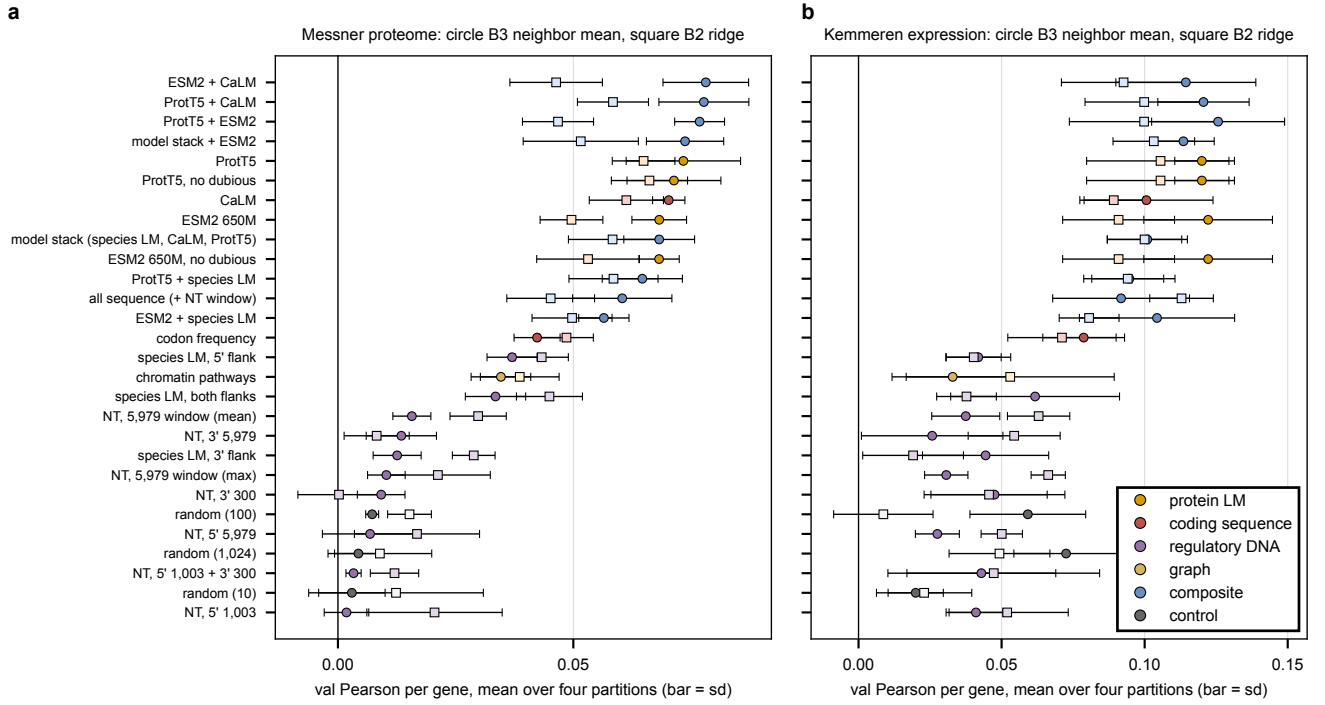
**The proteome floor is 0.07 to 0.08, and it is a protein-sequence floor.** On the proteome B3 reads 0.078 for ESM2 with CaLM and for ProtT5 with CaLM, 0.077 for ProtT5 with ESM2, 0.073 for ProtT5 alone, 0.070 for CaLM and 0.068 for ESM2; B2 peaks at 0.065 on ProtT5. The two protein language models are level, and every composite lands within 0.005 of its best member. Below them codon frequency reads 0.042, the species language model over both flanks 0.033, the chromatin-pathway vector 0.035, and every Nucleotide Transformer window 0.002 to 0.016 against random controls at 0.003 to 0.007. The v14 arms at their matched epoch read 0.114, 0.083, 0.112 and 0.085 on the four partitions (section 3.8), so the trained model's margin over the best linear floor is 0.00 to 0.04, and the floor was not raised by any representation the gate had missed.

**Expression reads the same order at twice the level.** B3 on ProtT5 reads  $0.120 \pm 0.009$  on validation and 0.118 on test, ESM2  $0.122 \pm 0.022$  and 0.093, ProtT5 with ESM2  $0.126 \pm 0.023$  and 0.107, CaLM 0.101 and 0.100; no composite beats ProtT5 on test. The all-sequence composite (7,168 dimensions) reads the highest B2 validation of the study, 0.113, and 0.099 on test, the width buying validation fit and not transfer. The regulatory-DNA representations read 0.03 to 0.06 on validation and 0.01 to 0.05 on test, where the 1,024-wide random control reads 0.073 on validation and 0.018 on test; the neighbor rule in a random space fits the validation draw through its choice of  $k$  and transfers nothing, which is the size of the selection optimism on this panel.

**What this settles for the model.** The input the trained model consumes, the four-embedding stack, is not a better perturbation representation than ProtT5 alone under either linear rule on either panel; the species-language-model flanks it carries are at the random floor by themselves. ESM2 is a substitute for ProtT5, not an upgrade, and adding it to the stack changes nothing a linear rule can see. The DNA models do not carry the deletion response under these rules, on either panel, at any window. Whether the network extracts something from the flanks that a bilinear map cannot is the embedding factor's question (section 3.1); the linear answer is that there is nothing there to extract at first order.

**Table 20:** Every gene representation the builder serves, as the perturbation input of the two embedding baselines on the four split partitions of the proteome and of the expression build. B2 is the bilinear ridge from the deleted gene’s embedding (rank and ridge selected on val), B3 the mean profile of the  $k$  nearest deleted genes in embedding space ( $k$  selected on val). Val is the mean  $\pm$  sd over the four seeds of the per-gene Pearson; test is the mean at the val-selected cell. Ordered by the proteome’s B3 val mean. SRC: experiments/019-simb-multimodal/scripts/baselines\_embedding\_study.py

| representation                         | family          | dim   | proteome          |        |                   |       | expression        |       |                   |       |
|----------------------------------------|-----------------|-------|-------------------|--------|-------------------|-------|-------------------|-------|-------------------|-------|
|                                        |                 |       | B2 val            | test   | B3 val            | test  | B2 val            | test  | B3 val            | test  |
| ESM2 + CaLM                            | composite       | 2,048 | 0.046 $\pm$ 0.010 | 0.053  | 0.078 $\pm$ 0.009 | 0.075 | 0.093 $\pm$ 0.022 | 0.099 | 0.114 $\pm$ 0.024 | 0.104 |
| ProtT5 + CaLM                          | composite       | 1,792 | 0.058 $\pm$ 0.008 | 0.061  | 0.078 $\pm$ 0.010 | 0.081 | 0.100 $\pm$ 0.021 | 0.106 | 0.121 $\pm$ 0.016 | 0.108 |
| ProtT5 + ESM2                          | composite       | 2,304 | 0.047 $\pm$ 0.008 | 0.051  | 0.077 $\pm$ 0.005 | 0.074 | 0.100 $\pm$ 0.026 | 0.104 | 0.126 $\pm$ 0.023 | 0.107 |
| model stack + ESM2                     | composite       | 4,608 | 0.052 $\pm$ 0.012 | 0.050  | 0.074 $\pm$ 0.008 | 0.072 | 0.103 $\pm$ 0.014 | 0.098 | 0.114 $\pm$ 0.011 | 0.091 |
| ProtT5                                 | protein LM      | 1,024 | 0.065 $\pm$ 0.007 | 0.062  | 0.073 $\pm$ 0.012 | 0.078 | 0.106 $\pm$ 0.026 | 0.110 | 0.120 $\pm$ 0.009 | 0.118 |
| ProtT5, no dubious                     | protein LM      | 1,024 | 0.066 $\pm$ 0.008 | 0.056  | 0.071 $\pm$ 0.010 | 0.079 | 0.106 $\pm$ 0.026 | 0.110 | 0.120 $\pm$ 0.009 | 0.118 |
| CaLM                                   | coding sequence | 768   | 0.061 $\pm$ 0.008 | 0.064  | 0.070 $\pm$ 0.003 | 0.070 | 0.089 $\pm$ 0.010 | 0.092 | 0.101 $\pm$ 0.023 | 0.100 |
| ESM2 650M                              | protein LM      | 1,280 | 0.050 $\pm$ 0.007 | 0.056  | 0.068 $\pm$ 0.006 | 0.062 | 0.091 $\pm$ 0.020 | 0.101 | 0.122 $\pm$ 0.022 | 0.093 |
| model stack (species LM, CaLM, ProtT5) | composite       | 3,328 | 0.058 $\pm$ 0.009 | 0.055  | 0.068 $\pm$ 0.008 | 0.067 | 0.100 $\pm$ 0.013 | 0.089 | 0.101 $\pm$ 0.014 | 0.091 |
| ESM2 650M, no dubious                  | protein LM      | 1,280 | 0.053 $\pm$ 0.011 | 0.057  | 0.068 $\pm$ 0.004 | 0.061 | 0.091 $\pm$ 0.020 | 0.101 | 0.122 $\pm$ 0.022 | 0.093 |
| ProtT5 + species LM                    | composite       | 2,560 | 0.059 $\pm$ 0.009 | 0.052  | 0.065 $\pm$ 0.009 | 0.058 | 0.094 $\pm$ 0.013 | 0.083 | 0.095 $\pm$ 0.016 | 0.077 |
| all sequence (+ NT window)             | composite       | 7,168 | 0.045 $\pm$ 0.009 | 0.048  | 0.060 $\pm$ 0.011 | 0.055 | 0.113 $\pm$ 0.011 | 0.099 | 0.092 $\pm$ 0.024 | 0.080 |
| ESM2 + species LM                      | composite       | 2,816 | 0.050 $\pm$ 0.008 | 0.051  | 0.056 $\pm$ 0.005 | 0.052 | 0.081 $\pm$ 0.010 | 0.084 | 0.104 $\pm$ 0.027 | 0.083 |
| codon frequency                        | coding sequence | 64    | 0.049 $\pm$ 0.006 | 0.042  | 0.042 $\pm$ 0.005 | 0.043 | 0.071 $\pm$ 0.019 | 0.052 | 0.079 $\pm$ 0.014 | 0.062 |
| species LM, 5’ flank                   | regulatory DNA  | 768   | 0.043 $\pm$ 0.006 | 0.037  | 0.037 $\pm$ 0.005 | 0.034 | 0.040 $\pm$ 0.010 | 0.041 | 0.042 $\pm$ 0.011 | 0.033 |
| chromatin pathways                     | graph           | 197   | 0.039 $\pm$ 0.008 | 0.046  | 0.035 $\pm$ 0.006 | 0.031 | 0.053 $\pm$ 0.036 | 0.020 | 0.033 $\pm$ 0.021 | 0.002 |
| species LM, both flanks                | regulatory DNA  | 1,536 | 0.045 $\pm$ 0.007 | 0.036  | 0.033 $\pm$ 0.006 | 0.032 | 0.038 $\pm$ 0.010 | 0.040 | 0.062 $\pm$ 0.030 | 0.048 |
| NT, 5,979 window (mean)                | regulatory DNA  | 2,560 | 0.030 $\pm$ 0.006 | 0.021  | 0.016 $\pm$ 0.004 | 0.008 | 0.063 $\pm$ 0.011 | 0.049 | 0.037 $\pm$ 0.012 | 0.017 |
| NT, 3’ 5,979                           | regulatory DNA  | 2,560 | 0.008 $\pm$ 0.007 | 0.008  | 0.013 $\pm$ 0.007 | 0.003 | 0.054 $\pm$ 0.016 | 0.023 | 0.026 $\pm$ 0.025 | 0.011 |
| species LM, 3’ flank                   | regulatory DNA  | 768   | 0.029 $\pm$ 0.005 | 0.017  | 0.013 $\pm$ 0.005 | 0.002 | 0.019 $\pm$ 0.018 | 0.021 | 0.044 $\pm$ 0.022 | 0.021 |
| NT, 5,979 window (max)                 | regulatory DNA  | 2,560 | 0.021 $\pm$ 0.011 | 0.020  | 0.010 $\pm$ 0.004 | 0.008 | 0.066 $\pm$ 0.006 | 0.060 | 0.031 $\pm$ 0.008 | 0.024 |
| NT, 3’ 300                             | regulatory DNA  | 2,560 | 0.000 $\pm$ 0.009 | 0.001  | 0.009 $\pm$ 0.005 | 0.003 | 0.046 $\pm$ 0.020 | 0.029 | 0.048 $\pm$ 0.025 | 0.016 |
| random (100)                           | control         | 100   | 0.015 $\pm$ 0.005 | 0.010  | 0.007 $\pm$ 0.001 | 0.003 | 0.009 $\pm$ 0.017 | 0.011 | 0.059 $\pm$ 0.020 | 0.037 |
| NT, 5’ 5,979                           | regulatory DNA  | 2,560 | 0.017 $\pm$ 0.013 | -0.018 | 0.007 $\pm$ 0.010 | 0.005 | 0.050 $\pm$ 0.007 | 0.008 | 0.028 $\pm$ 0.008 | 0.021 |
| random (1,024)                         | control         | 1,024 | 0.009 $\pm$ 0.011 | -0.004 | 0.004 $\pm$ 0.005 | 0.001 | 0.049 $\pm$ 0.018 | 0.013 | 0.073 $\pm$ 0.018 | 0.018 |
| NT, 5’ 1,003 + 3’ 300                  | regulatory DNA  | 5,120 | 0.012 $\pm$ 0.005 | 0.009  | 0.003 $\pm$ 0.002 | 0.005 | 0.047 $\pm$ 0.037 | 0.041 | 0.043 $\pm$ 0.026 | 0.045 |
| random (10)                            | control         | 10    | 0.012 $\pm$ 0.019 | 0.006  | 0.003 $\pm$ 0.007 | 0.001 | 0.023 $\pm$ 0.017 | 0.003 | 0.020 $\pm$ 0.010 | 0.002 |
| NT, 5’ 1,003                           | regulatory DNA  | 2,560 | 0.021 $\pm$ 0.014 | 0.008  | 0.002 $\pm$ 0.005 | 0.005 | 0.052 $\pm$ 0.021 | 0.044 | 0.041 $\pm$ 0.009 | 0.016 |



**Figure 17: Protein and coding-sequence representations carry the deleted gene’s response on both panels, no protein language model beats ProtT5, composites add nothing the seed spread can resolve, and every regulatory-DNA representation sits at the random floor. a, the Messner proteome; b, Kemmeren expression.** Each row is one representation as the perturbation input of B3, the neighbor mean (circle), and B2, the bilinear ridge (square); the point is the mean validation per-gene Pearson over the four split partitions and the bar its sd. Rows ordered by the proteome’s B3 mean; colors by representation family. Generated by `baselines_embedding_study.py`.

## 5 What the evidence leaves open, and what fits the hardware x

Each item is a decision the measurements above do not make, stated with the measurement that would. The hardware they have to fit, as of 2026-09-09 18:10 CT: on IGB, all eight cabbi RTX 6000 cards are allocated (three to another user, five to this campaign’s surviving runs), the two batch-64 Pearson tasks free two cards when they reach 9,900 in about six hours, the packed anchored-Pearson task frees one about a day later, and the ranking round holds two cabbi cards and two A40s for one to two more days; four A40s on the `gpu` partition are idle; the `mmli` A100s are not to be touched. On Delta the `bbub` allocation holds 3,738 GPU-hours under a 48-hour wall, which at the `v10` grid’s measured rate of 990 to 1,400 epochs per task fits every 1,000-epoch design below and none of the 9,900-epoch ones.

**Which number is the target.** Stopping on the validation loss says every run is done by epoch 300 to 700 and the long-budget campaign is over; stopping on Pearson says the curves have not turned at 19,000 (section 2.6). The manuscript reports Pearson and the objective is the pinball loss, and the two metric-aligned rounds show the naive resolution is unstable at batch 32 under Pearson, no better than the incumbent at batch 64, and slow under ListMLE (section 3.3, section 3.4). The decision is about what the paper claims, not about what to run.

**The incumbent’s embedding against ProtT5: the cheapest contrast on the table, and it fits Delta.** The grid measured content against random at width 1,024 and did not include `calm` (section 3.1). A two-arm replicate set at the incumbent configuration, `calm` against `prot_T5_all`, three seeds each at 1,000 epochs, resolves a gap of about 0.03 at the within-cell replicate sd of 0.012 to 0.025, and is six runs of under 48 hours each: one Delta array on `bbub`, about 300 GPU-hours. It also gives the first true replicate spread of the incumbent configuration, which the eight long-budget arms only bound, and that spread is what every later design is sized by.

**The per-gene readout at a resolving replicate count, also on Delta.** +0.027 at two pairs against a resolution of 0.06 (section 3.2). Detecting 0.03 at the 4,000-epoch spread of 0.0171 needs about six pairs; at 1,000 epochs and 0.0099 it needs three, which is six runs, `R_ref` against `R_pergene`, and a second 48-hour Delta array. Whether the arm’s early peak and give-back survive a shorter budget is what the same runs show.

**The readout round, submitted 2026-09-10 on IGB.** The published readouts reduce to a short list once the shared trunk is fixed: GEARS’s gene-specific row on the gene’s own token, that row plus GEARS’s pooled cross-gene state, State’s  $\mathbf{W}_{\text{recon}}$  and the benchmark’s ridge decoder (a gene-specific row on the strain vector), `scGPT`’s masked-value form (a factored bilinear, gene sensitivity times strain amplitude), State’s embedding-model decoder (a shared

MLP over the gene embedding and the cell vector), and the shared linear map. Each maps to one existing option of the per-gene head except the strain-vector row, added as `context_readout` and zero-gated like the others. Eight arms by four seeds, four runs per 48 GB card (a batch-32 run peaks near 9 GB on the W&B system metrics), seed-major so every contrast against the shared-MLP reference is paired within seed with the card as a block, on the full-locus input of section 5's embedding round under the pinball objective, at that round's 1,400 epochs so the reference arm replicates its widest arm: jobs 2392379 (cabbi) and 2392381 (gpu), source 271b2c71. The wiring of `linear_readout` was found broken while building it (section 3.1), so the linear arm is its first measurement. Four pairs per contrast at a paired sd near 0.03 resolve about 0.035; the full-locus input is itself unmeasured against `calm` until the embedding round lands, and every contrast here is within it.

**The batch-32 solo Pearson run that separates batch size from placement.** The batch-64 pair are the round's only informative survivors and batch size is confounded with solo placement in that arm (section 3.3). One batch-32 solo seed of the pure objective is the run that separates the two; it needs the 9,900-epoch budget, so it takes one cabbi card for 3.5 days when the batch-64 tasks free theirs.

**The ranking round, when it ends.** The four surviving runs reach 6,000 epochs in one to two days. The readout then is the matched-budget Spearman and Pearson against the long-budget arms at 6,000 ( $0.192 \pm 0.018$ ) and against the Pearson-objective survivors, and the one follow-up the current curves motivate is a learning-rate arm, since the slow start is the only thing separating ListMLE from the band so far and the rate was inherited from the pinball loss. That is a hypothesis until the round reads out.

**The perturbation never touches the encoder, and the CLS state is wild-type.** This is the structural fact of section 1.2: the encoder runs once on the unperturbed graph,  $h_{\text{CLS}}$  is byte-identical across strains, and the deletion is applied to finished gene states through a one-key cross-attention. Every mechanism arm so far has been a different way of applying the perturbation after the encoder, and none puts it inside. Three designs would, none of them trained here, listed by how much they change:

- **Perturb the CLS as well as the genes.** Run the existing perturbation operator on  $h_{\text{CLS}}$  too, cross-attending it to the deleted set, so the whole-cell summary becomes strain-specific at the cost of one more query. Cheapest, restores a strain-dependent global state, and leaves the pair-term degeneracy exactly where it is.
- **Edit the token before the encoder.** Remove or flag the deleted gene's token in the encoder input (Geneformer's in-silico deletion, scGPT's perturbation flag), so the graph-masked self-attention carries the deletion outward to every gene and to CLS along the interaction graphs, and the one-key degeneracy never arises. This is the design that uses the graphs for the perturbation rather than only for the wild-type state. Its cost is one encoder pass per strain instead of one per batch; hypothesis (untested), at 39 batches per epoch and 17 to 31 s/epoch that is a wall-clock multiplier near the batch size unless a shallow strain-specific encoder is stacked on frozen wild-type layer outputs.
- **Perturbation tokens, typed by what is known.** The token-edit design works because a deletion names its node. An environmental perturbation, an antifungal added to the medium, names no gene, so it cannot be an edit; the generic form is a set of typed perturbation tokens entering the encoder beside the gene tokens, a gene-edit token that attends to its gene and an environment token that attends to CLS or to every gene, with the encoder deciding which genes an environment token reaches. That covers deletions and environments in one mechanism and makes CLS strain-specific by construction, and it is the same encoder-per-strain cost as the token edit.

None of these is launchable this week: the first is a small change to the model file, the other two restructure the forward pass, and all three are hypotheses about mechanism until run at a matched budget against the incumbent.

**Packing.** Four of four packed five-day tasks died of host memory (section 3.6). Until the growth is identified, a five-day arm is either solo on its card or budgeted at 48 GB per run rather than 30, and either choice halves the slots the launch arithmetic assumed. Delta's 48-hour wall sidesteps it for the short designs.

**The per-gene mean on squared error.** No live run beats  $\text{nmse} = 1$  at its Pearson peak (section 2.1). The one-parameter rescale argument of section 1.2's source, multiplying every prediction by  $r/s$  estimated on train, changes no correlation and takes  $\text{nmse}$  to  $1 - r^2$ ; it has not been applied to any run of the campaign, costs nothing, and belongs at the point a figure is drawn.

**The proteome round, submitted 2026-09-15 on cabbi.** The split round's design on the Messner 2023 knockout proteome (section 3.8): `P_ref` and `P_concat` on split seeds 0 to 3, two initialization seeds, four runs per RTX 6000 card, 2,000 epochs, job 2400350 (`cgt_expr_v14_proteome.yaml`). The build `fig3_proteome` unions Kemmeren, Sameith and Messner in one genotype-keyed store; the proteome label is  $\log_2$  of the strain over the 388-replicate HIS3 reference, converted at build time, over the union of 1,850 proteins with a strain's unmeasured proteins carried as missing and excluded from every loss and metric (strains quantify 1,441 to 1,850). The per-gene head reads that label unchanged

and logs under `val/proteome`. The linear baselines on the same four partitions read 0.058 to 0.074 for the bilinear ridge on ProtT5 and 0.057 to 0.083 for the neighbor mean (validation), against a ceiling of 0.40 to 0.70 (section 2.2); the round is read as v13 is, the paired contrast at a matched epoch and the test score at the best-validation checkpoint. Nothing about it is a result until the runs finish. SRC: `experiments/019-simb-multimodal/results/baselines_split_fig3_proteome/seed0.jsontoseed3.json`